

MyBudget

Gestión de presupuesto familiar

Trabajo Fin de Máster

Alejandro Rafael Alfaro Soto

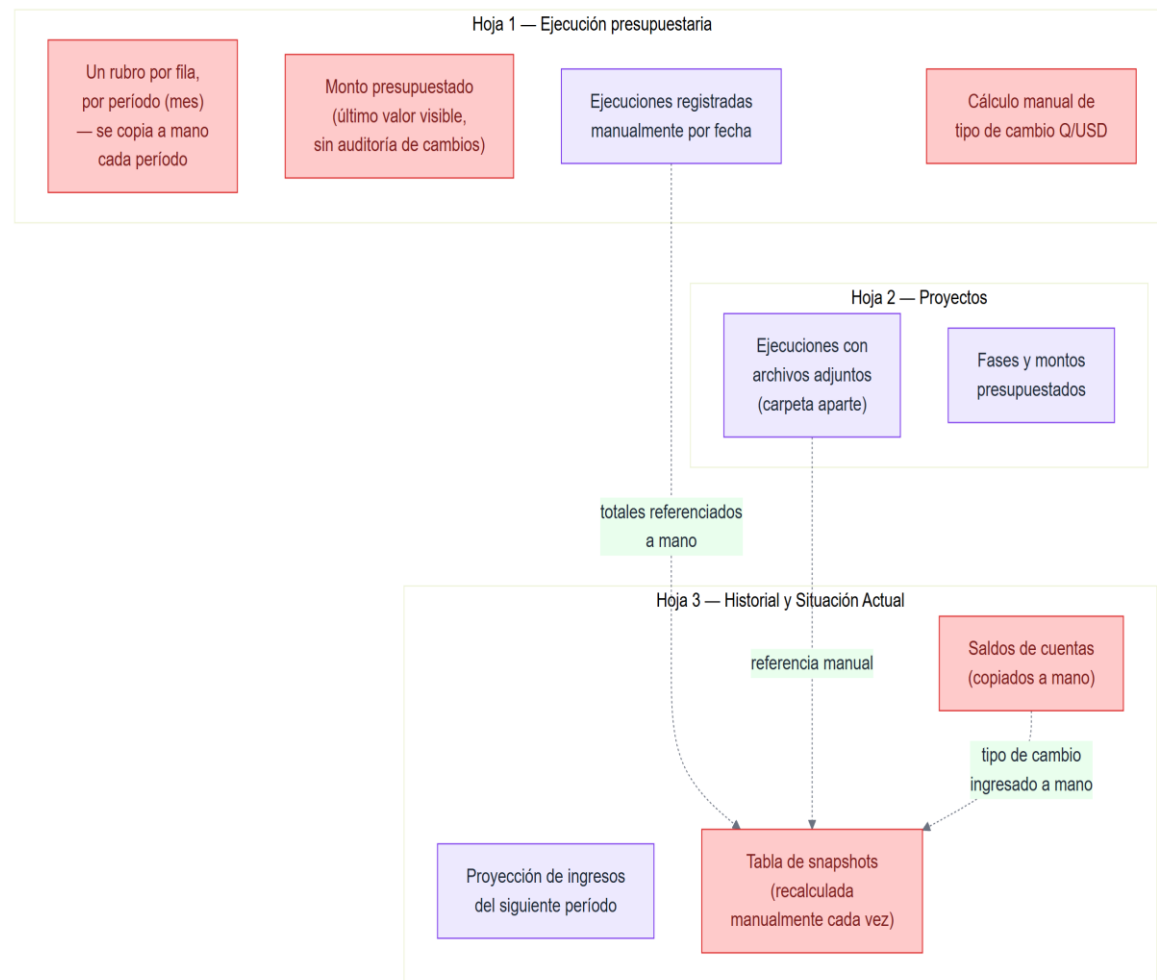
Master de Desarrollo con IA

2026

CONTEXTO

El problema: la hoja de cálculo

- Presupuesto familiar gestionado en Excel con 3 hojas: Ejecución presupuestaria, Proyectos, e Historial y Situación Actual.
- Un rubro por fila obliga a re-crear cada período manualmente.
- El historial de cambios de monto presupuestado se pierde o es difícil de rastrear.
- Cálculo manual de tipo de cambio Quetzal/Dólar en cada registro.
- Sin control de acceso: cualquiera con el archivo puede editar todo.



La propuesta

- Aplicación web responsive: Vue en el frontend, .NET en el backend, PostgreSQL como base de datos.
- Múltiples presupuestos por propietario, con roles: Owner, Admin, Operator, Read-only.
- Ciclos y períodos, rubros tipados (Gasto, Ahorro largo plazo, Ahorro preventivo).
- Snapshots de situación actual (el "corte") y gráficas comparativas.
- MVP A construido en este TFM; MVP B (proyectos, deudas/cuotas, import/export) queda diferido — alcance controlado.

Qué hace MyBudget (1/3)

Cuentas y acceso

- Múltiples presupuestos por usuario, invitación de otros usuarios por email.
- 4 roles (Owner/Admin/Operator/Read-only) → colaboración familiar sin perder control de quién edita qué.
- Login JWT, recuperación de contraseña, bloqueo tras intentos fallidos → seguridad sin fricción para uso diario.

Estructura de presupuesto

- Ciclos y períodos con tipo de cambio propio por ciclo → refleja presupuestos anuales/mensuales reales, multi-moneda.
- Rubros con historial de revisiones sin perder la auditoría → resuelve el dolor #1 de la hoja de cálculo.
- Categorías y grupos reordenables → organización flexible, no columnas fijas como en Excel.

Qué hace MyBudget (2/3)

Ejecución (gasto real)

- Matriz multi-período con CRUD en línea → registrar gasto real sin salir de la vista comparativa.
- Notas de crédito/débito, tipo de cambio por entrada → maneja casos reales que una hoja de cálculo no modela bien.

Situación actual

- Catálogo de cuentas bancarias + "corte" diario de saldos → saldo real vs. presupuestado/ejecutado, de un vistazo.
- Multi-moneda con tipo de cambio congelado (histórico) vs. transaccional (actual) → evita comparar cifras no comparables.

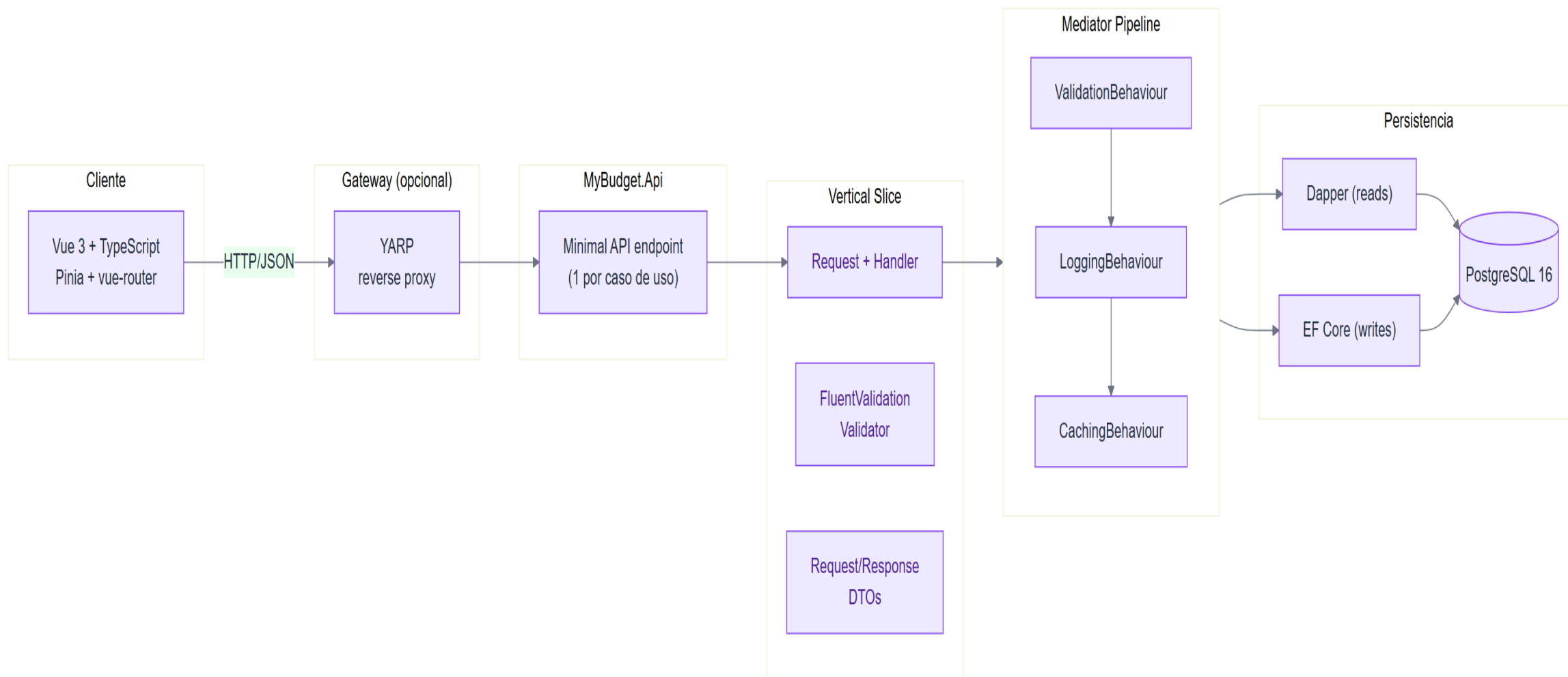
Qué hace MyBudget (3/3)

- Dashboard analítico: tendencia histórica, banda de comportamiento promedio, comparación de rubros por período/ciclo → convierte datos crudos en decisión.
- Auditoría completa de mutaciones + log de seguridad con retención de 90 días → trazabilidad total, no negociable al manejar dinero compartido.
- Localización completa ES/EN → usable por toda la familia, no solo por quien lo construyó.
- Fuera de alcance de este TFM (MVP B): proyectos, compromisos financieros, cuotas/deudas, import/export.

Stack tecnológico

Capa	Tecnología
Backend	.NET 10 · ASP.NET Core Minimal APIs · Mediator · Dapper + EF Core · FluentValidation
Frontend	Vue 3.5 (Composition API) · TypeScript · Pinia · Tailwind v4 + daisyUI · Chart.js
Base de datos	PostgreSQL 16
Pruebas	xUnit + NSubstitute (unit) · WebApplicationFactory (integration) · Vitest · Playwright (E2E)
Infraestructura	Docker Compose (Postgres, Redis, Mailpit, Seq, Jaeger)

Arquitectura — Vertical Slice Architecture



Cada caso de uso vive en Features/<Área>/<CasoDeUso>/ con 4 archivos: request+handler, validator, endpoint, DTOs.

Proceso de desarrollo — Spec-Driven Development

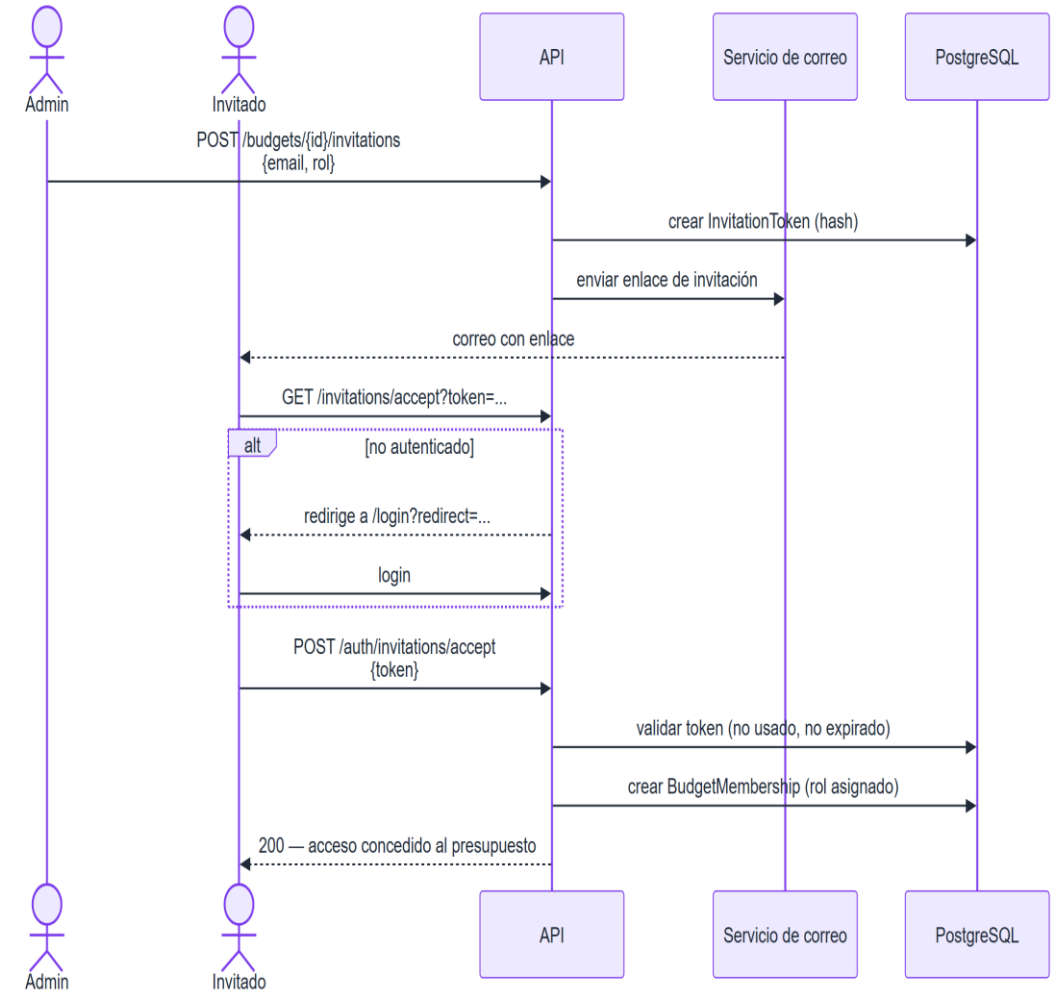
- Cada feature: exploración → propuesta → spec → diseño → tasks → implementación → verificación → archivo.
- 23 cambios documentados de extremo a extremo en [openspec/changes/archive/](#).
- Agentes de IA como herramienta dentro del proceso, dirigidos por el desarrollador — no autónomos.

Línea de tiempo — de 0 a MVP A

- 2026-07-07 — foundation: scaffold completo de la aplicación.
- 2026-07-08 — auth: registro, login, invitaciones, RBAC.
- 2026-07-10/11 — budget-structure + UI: ciclos, períodos, rubros, categorías.
- 2026-07-13/14 — budget-execution + UI: matriz de ejecución.
- 2026-07-29 — current-situation: cuentas bancarias y el "corte".
- 2026-08-04 — dashboard: última pieza de MVP A.
- Total: 23 cambios archivados, ~1 mes de desarrollo.

Cuentas y acceso

- Registro, login JWT (15 min) con refresh token rotativo (7 días).
- Invitación por email con rol asignado por el administrador.
- 4 roles por presupuesto: Owner, Admin, Operator, Read-only.
- Recuperación de contraseña por correo, con TTL configurable.



Registro — formulario

Create Account

First name

First name

Last name

Last name

Email

you@example.com

Password

Password (min 8 chars, 1 uppercase, 1 digit)

Language

English

Create Account

[Already have an account? Sign in](#)

Formulario vacío

Create Account

First name

E2E

Last name

Slides

Email

e2e-slide-register-1786313200333@example.com

Password

.....

✓ At least 8 characters

✓ One uppercase letter (A–Z)

✓ One lowercase letter (a–z)

✓ One number (0–9)

Language

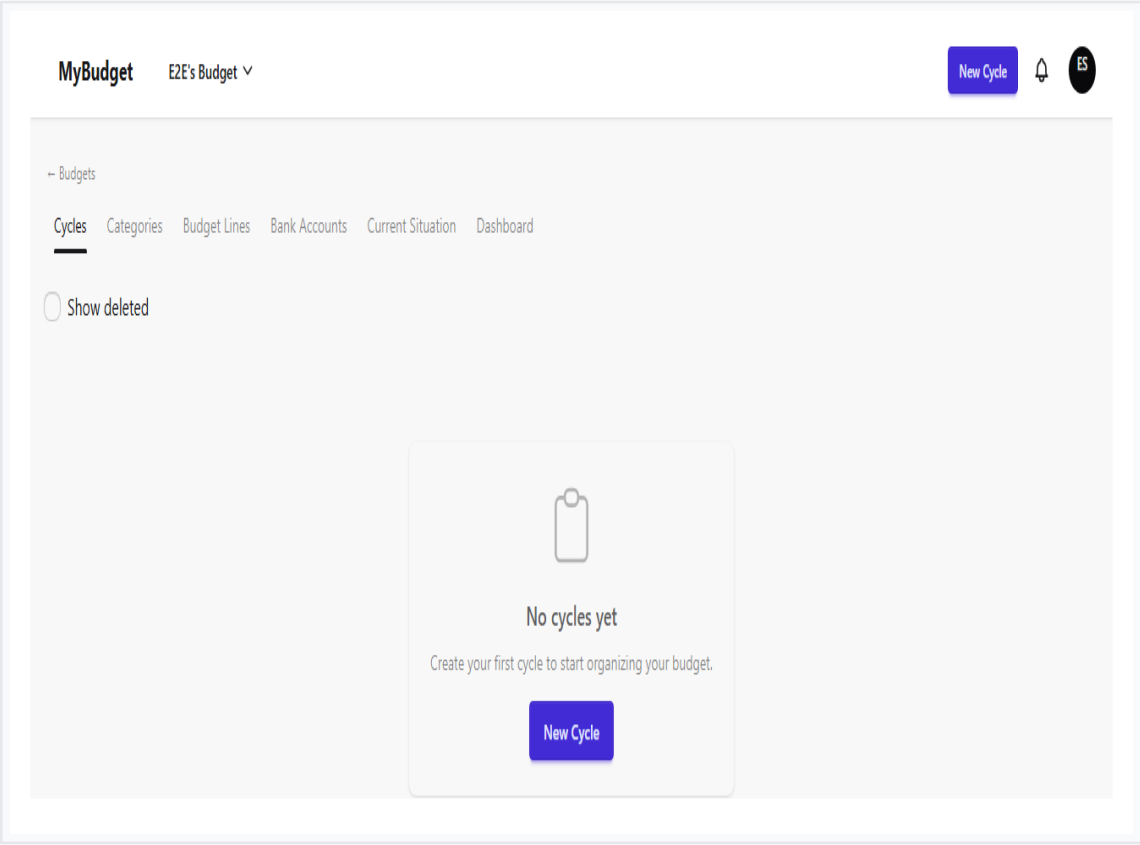
English

Create Account

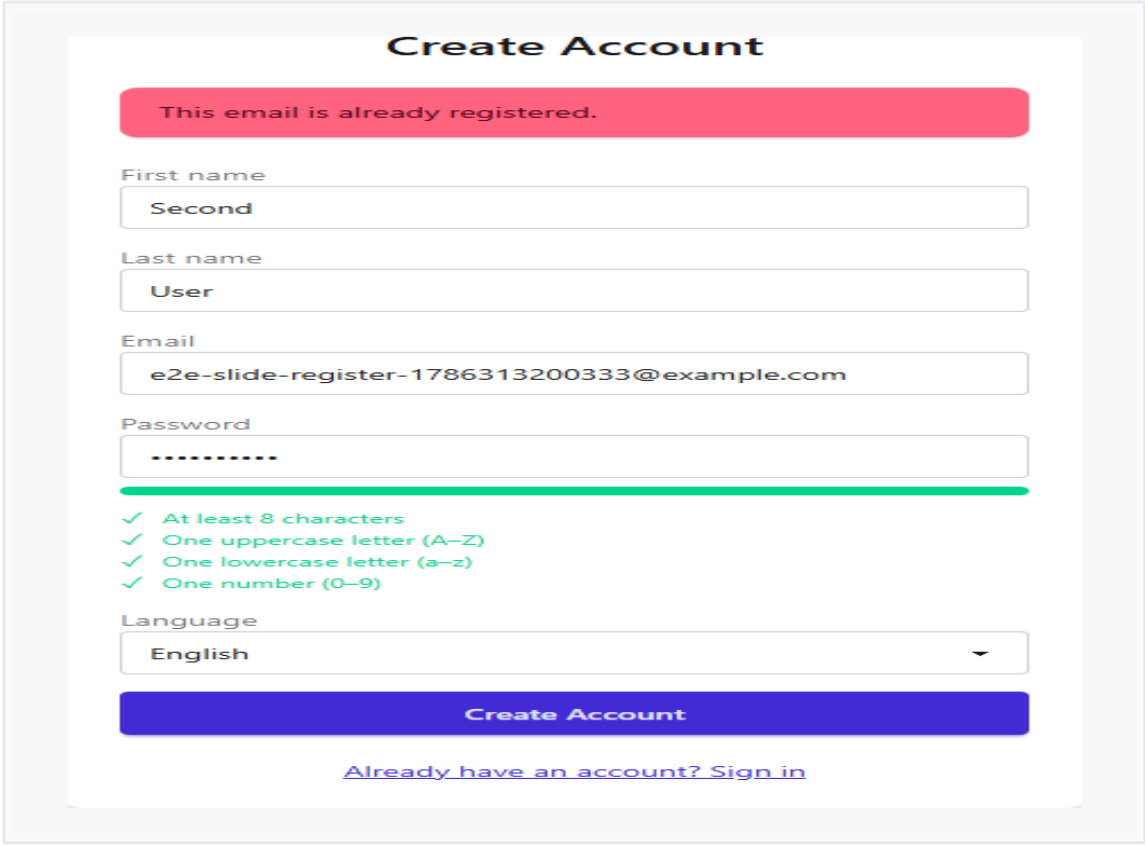
[Already have an account? Sign in](#)

Formulario completado

Registro — resultado



Registro exitoso



Error: correo duplicado

Login y Logout

Sign In

Email

you@example.com

Password

Password

Sign In

Forgot your password?

Don't have an account? Register

Login — vacío

MyBudgetEZE's Budget

New Cycle

Budgets

CyclesCategoriesBudget LinesBank AccountsCurrent SituationDashboard

Show deleted

No cycles yet

Create your first cycle to start organizing your budget.

New Cycle

Login exitoso

Sign In

Invalid email or password. Please try again.

Email

e2e-slide-login-1786313204667@example.com

Password

.....

Sign In

Forgot your password?

Don't have an account? Register

Credenciales inválidas

MyBudgetEZE's Budget

New Cycle

Budgets

CyclesCategoriesBudget LinesBank AccountsCurrent SituationDashboard

Show deleted

No cycles yet

Create your first cycle to start organizing your budget.

New Cycle

EZE Logout

e2e-slide-logout-0786313204667@example.com

Owner

Change password

ENES

Logout

Menú de usuario

Sign In

Email

you@example.com

Password

Password

Sign In

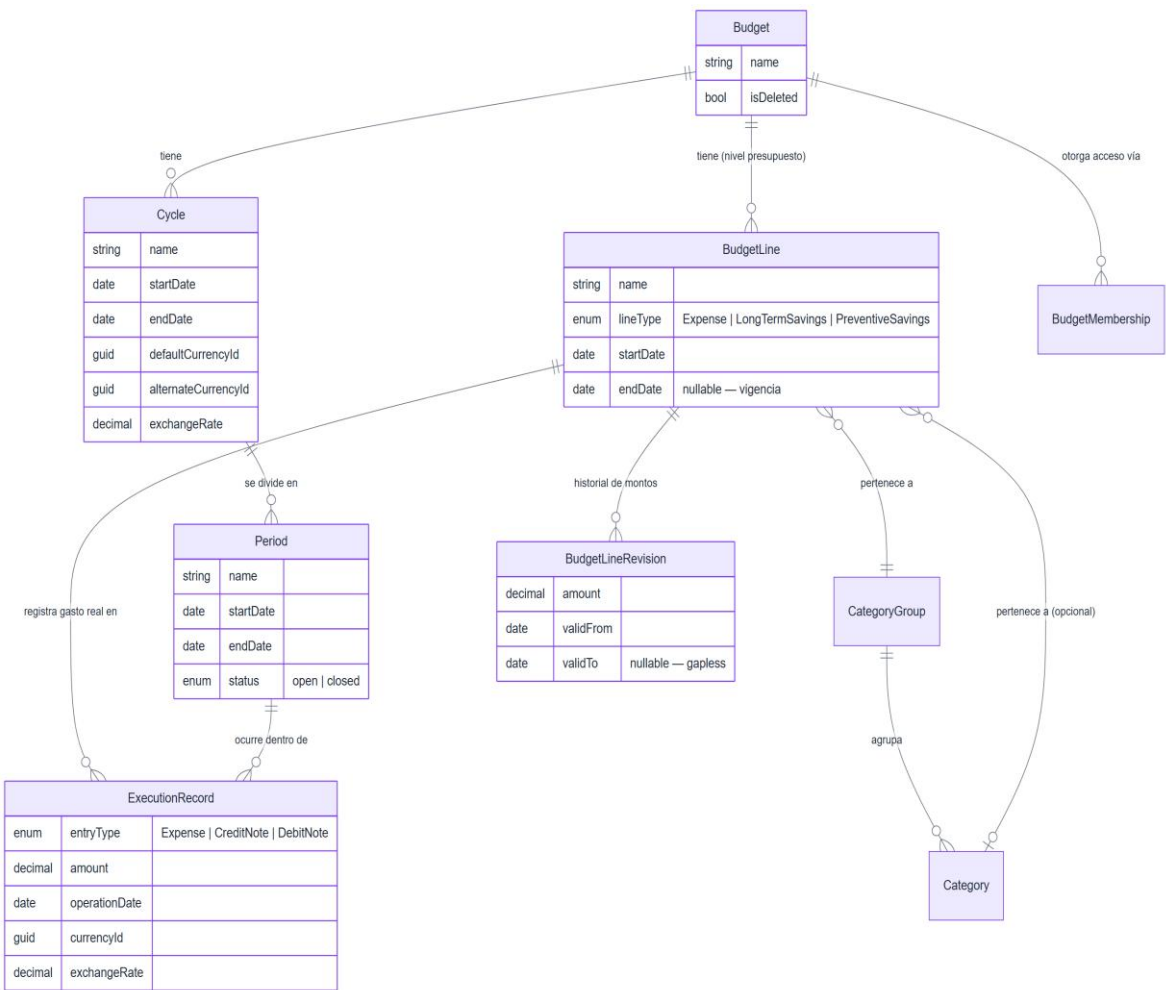
Forgot your password?

Don't have an account? Register

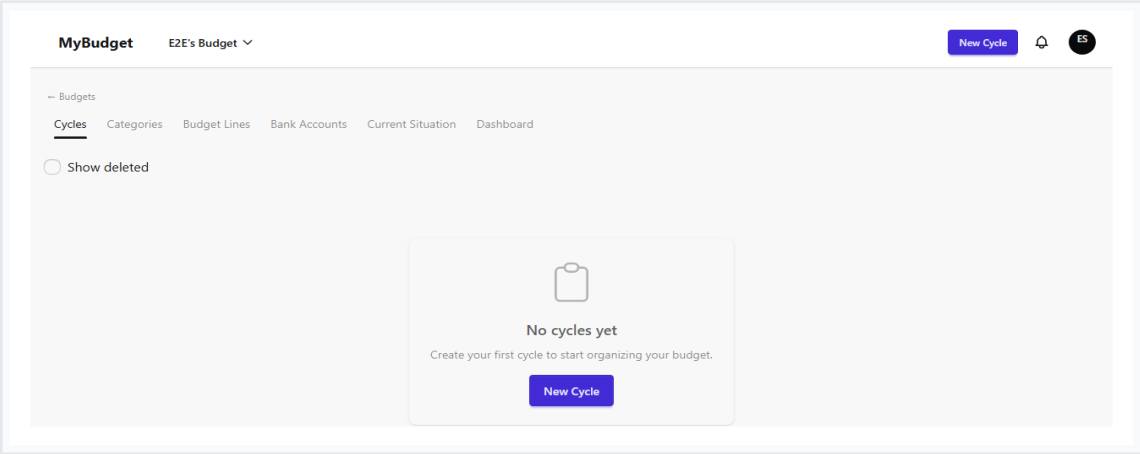
Logout exitoso

Estructura de presupuesto

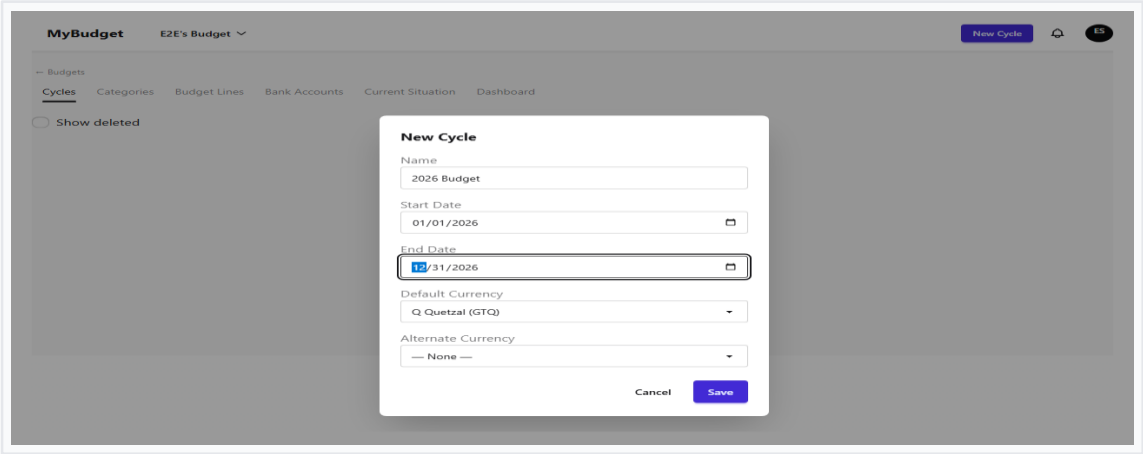
- Ciclos (ej. un año) divididos en Períodos (ej. meses).
- Rubros a nivel de presupuesto, con rango de vigencia propio (fecha inicio/fin).
- Historial de revisiones de monto (BudgetLineRevision) — sin huecos, sin perder el rastro de auditoría.
- Agrupados por CategoryGroup → Category.



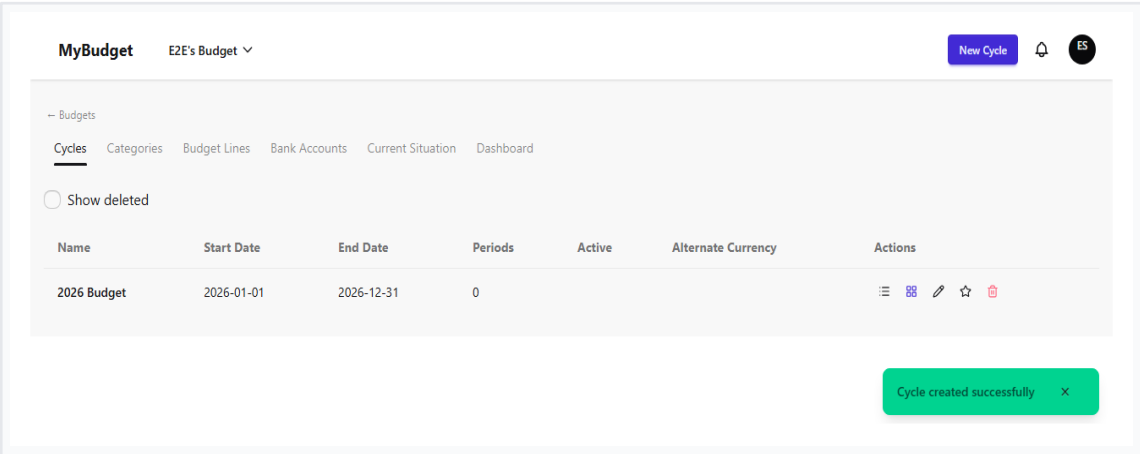
Ciclos — creación



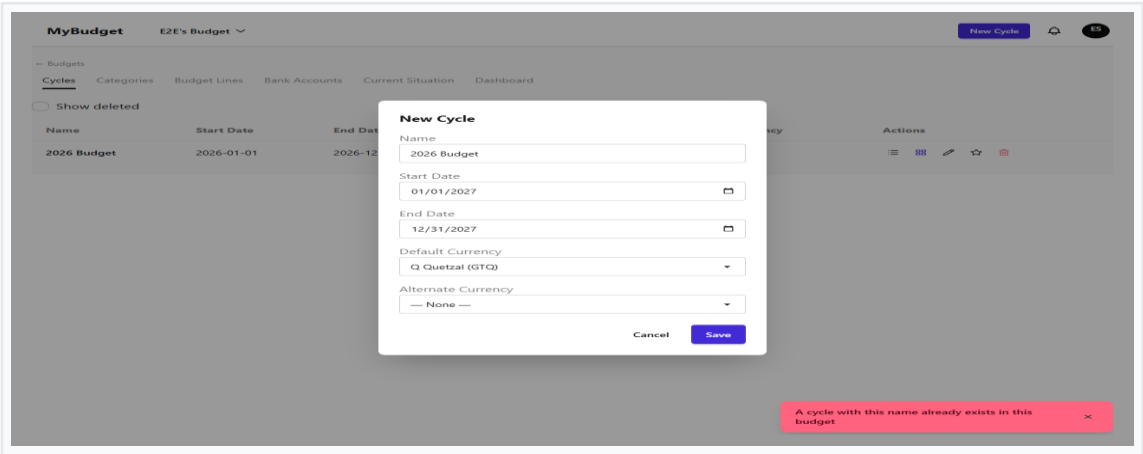
Lista vacía



Formulario de creación



Creación exitosa



Error: nombre duplicado

Ciclos — edición y ciclo de vida

MyBudgetEZE's Budget

New Cycle

Budgets

CyclesCategoriesBudget LinesBank AccountsCurrent SituationDashboard

Show deleted

Name	Start Date	End Date	Periods	Active	Alternate Currency	Actions
2026 Budget	2026-01-01	2026-12-31	0			

Edit Cycle

Name2026 Budget — Updated

Start Date01/01/2026

End Date12/31/2026

Default CurrencyQ Quetzal (GTQ)

Alternate CurrencyNone

CancelSave

Formulario de edición

MyBudgetEZE's Budget

New Cycle

Budgets

CyclesCategoriesBudget LinesBank AccountsCurrent SituationDashboard

Show deleted

Name	Start Date	End Date	Periods	Active	Alternate Currency	Actions
2026 Budget — Updated	2026-01-01	2026-12-31	0			

Cycle updated successfully

Edición exitosa

MyBudgetEZE's Budget

New Cycle

Budgets

CyclesCategoriesBudget LinesBank AccountsCurrent SituationDashboard

Show deleted

Name	Start Date	End Date	Periods	Active	Alternate Currency	Actions
2026 Budget — Updated	2026-01-01	2026-12-31	0	Active		

Cycle set as active

Ciclo activado

MyBudgetEZE's Budget

New Cycle

Budgets

CyclesCategoriesBudget LinesBank AccountsCurrent SituationDashboard

Show deleted

Name	Start Date	End Date	Periods	Active	Alternate Currency	Actions
2026 Budget — Updated	2026-01-01	2026-12-31	0	Active		

Delete Cycle

Are you sure you want to delete this cycle?

CancelConfirm

Confirmación de borrado

MyBudgetEZE's Budget

New Cycle

Budgets

CyclesCategoriesBudget LinesBank AccountsCurrent SituationDashboard

Show deleted

No cycles yet

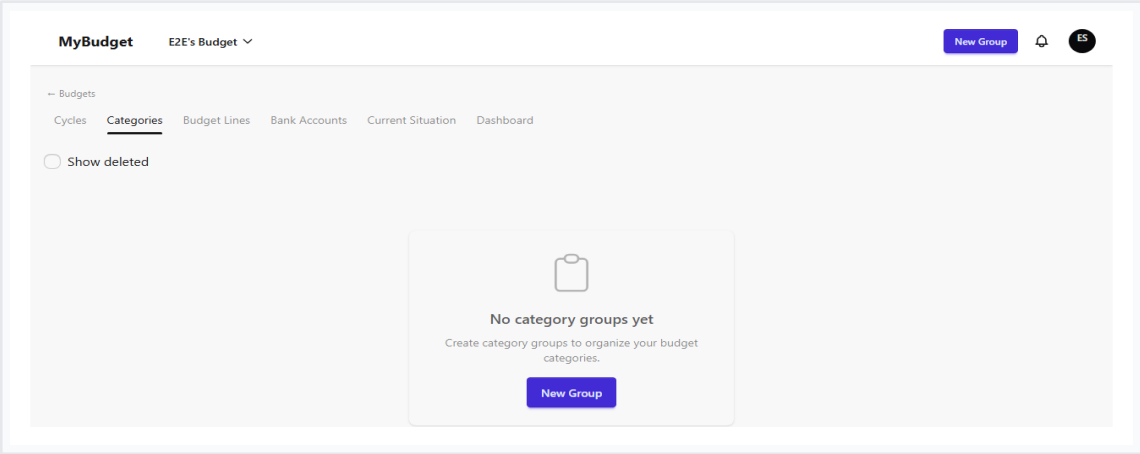
Create your first cycle to start organizing your budget.

New Cycle

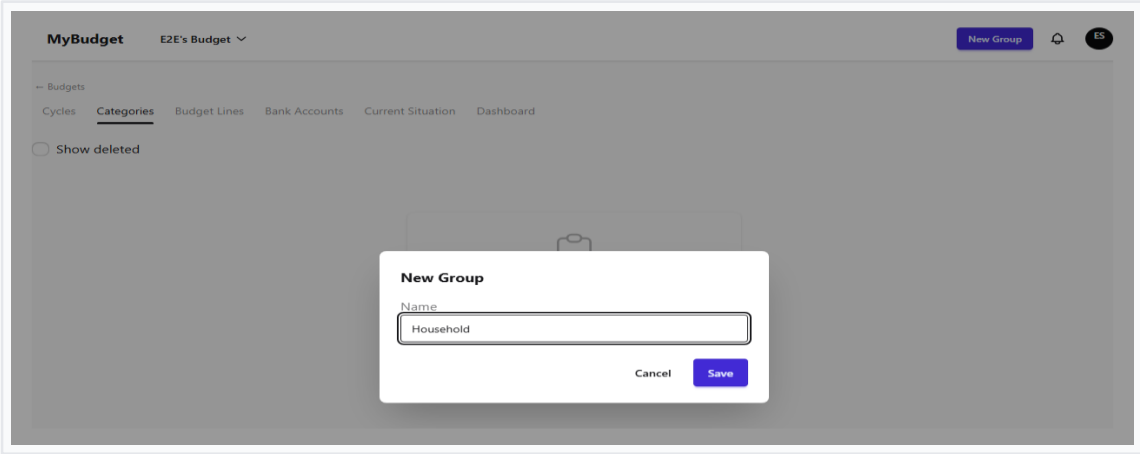
Cycle deleted successfully

Borrado exitoso

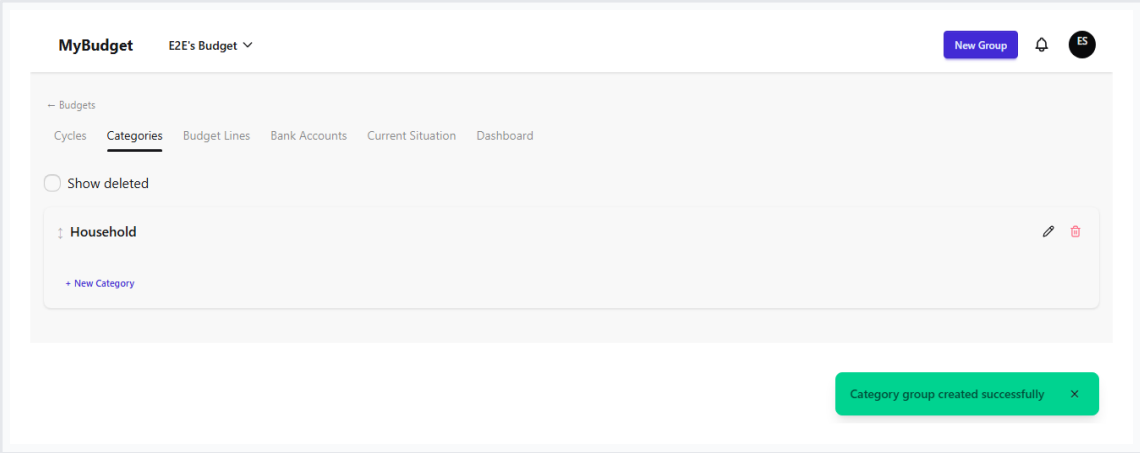
Categorías — grupos



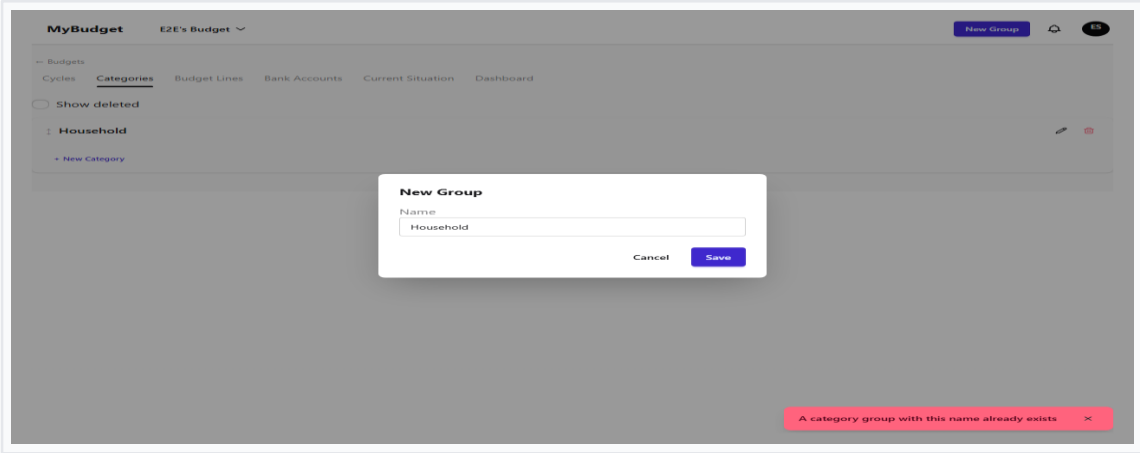
Lista vacía



Crear grupo

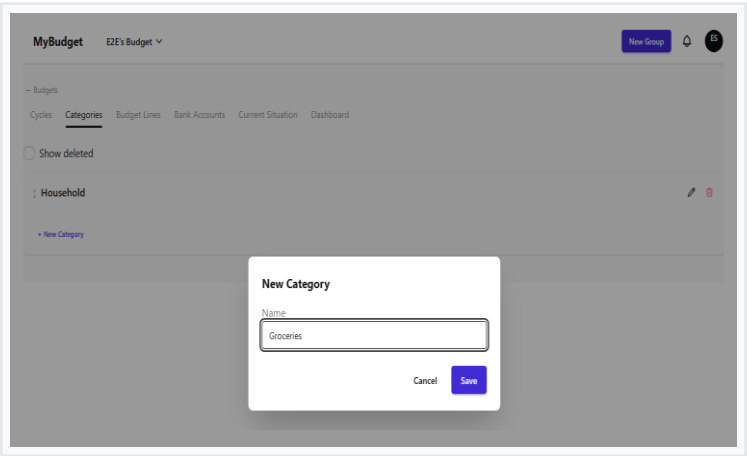


Creación exitosa

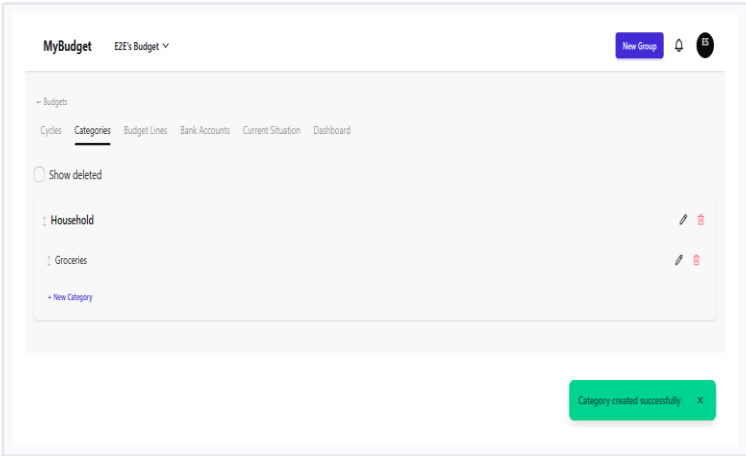


Error: nombre duplicado

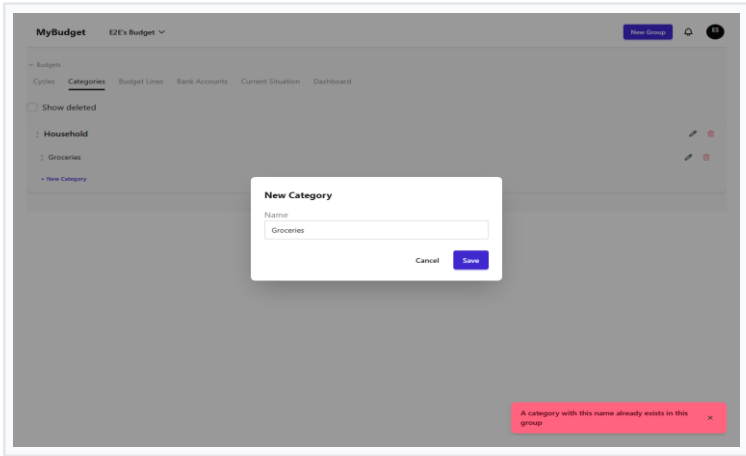
Categorías — categorías y ciclo de vida



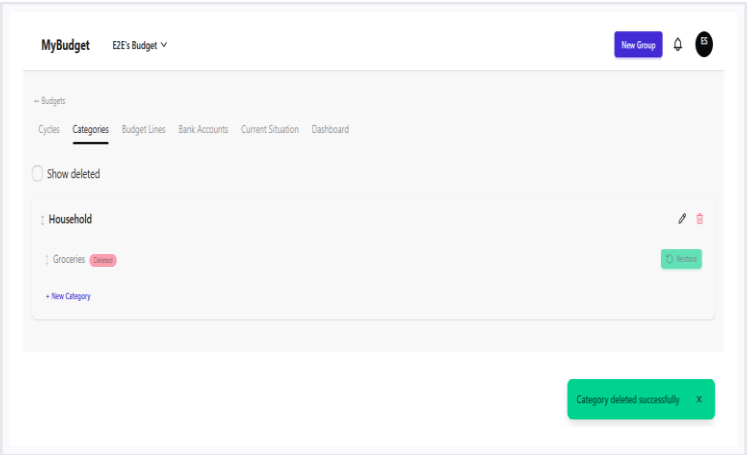
Crear categoría



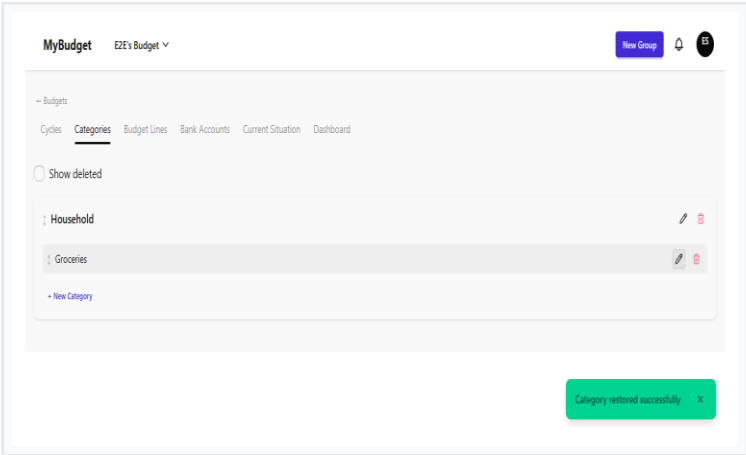
Creación exitosa



Error: nombre duplicado

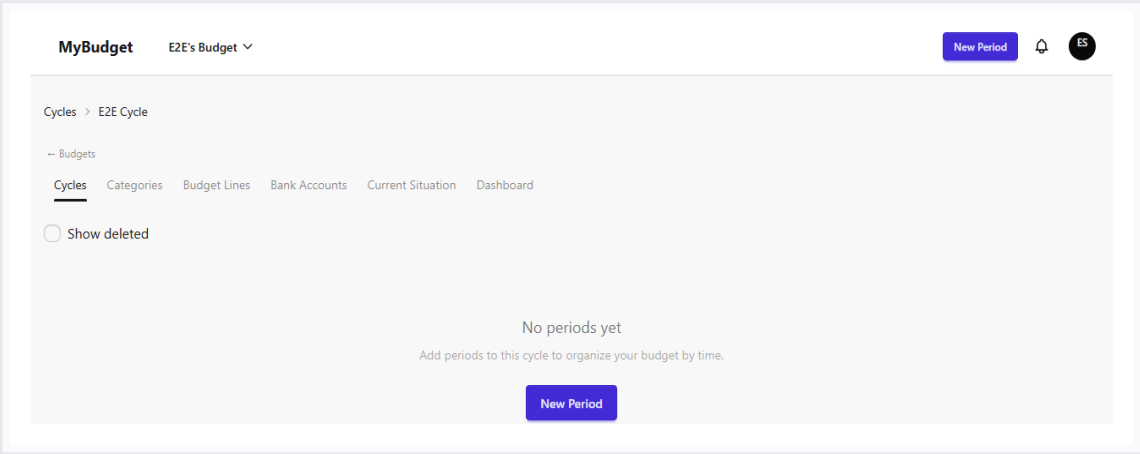


Borrado exitoso

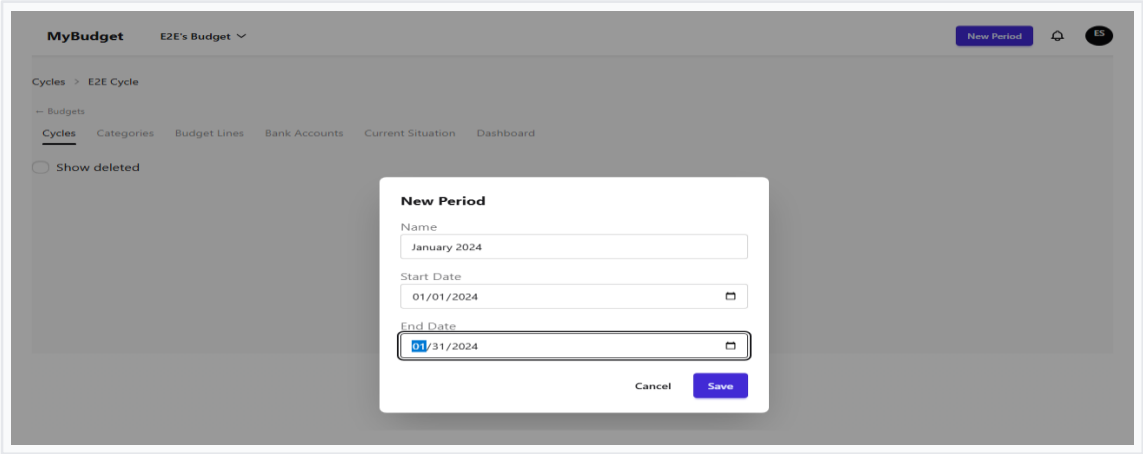


Restauración exitosa

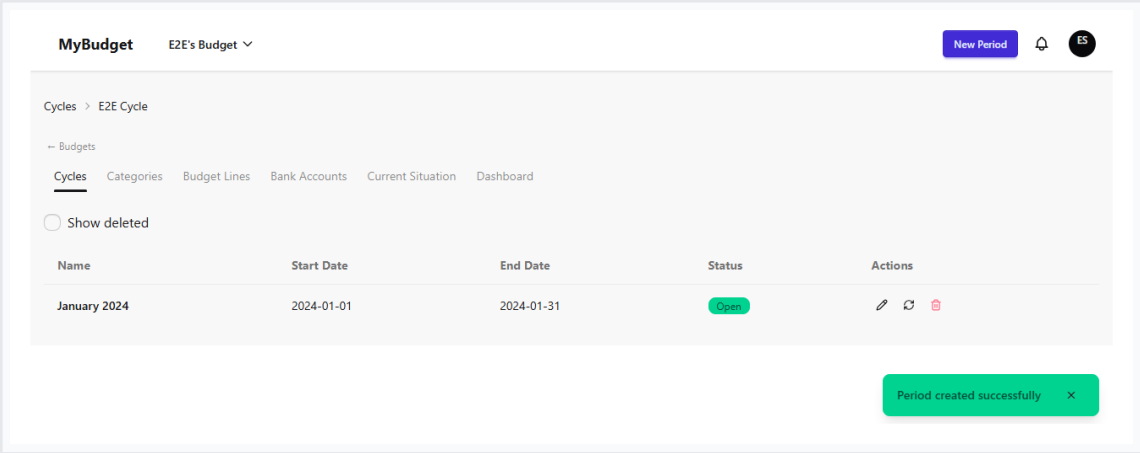
Períodos — creación



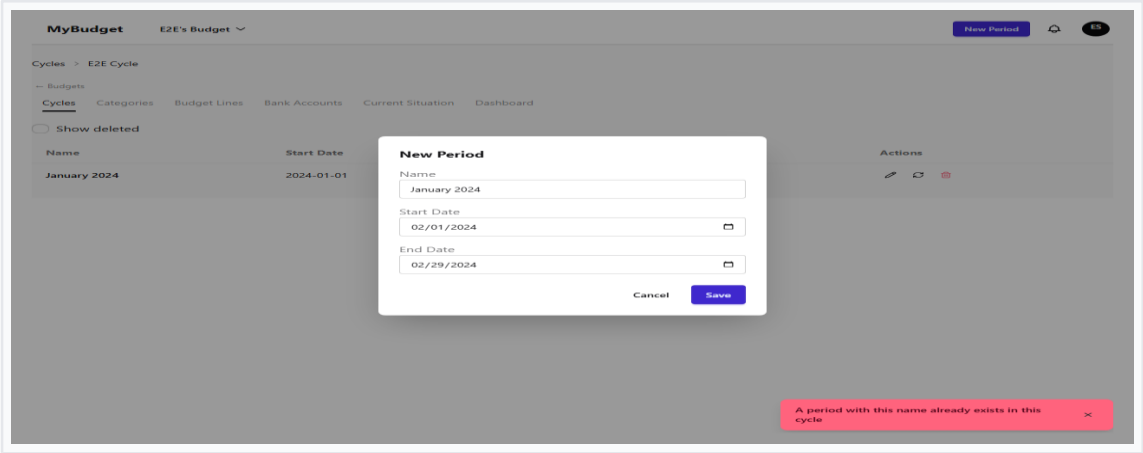
Lista vacía



Formulario de creación

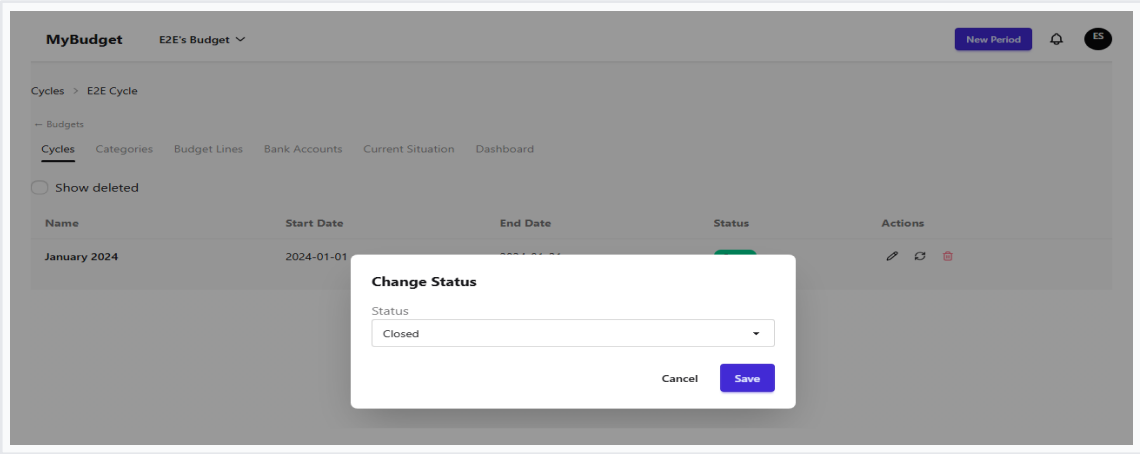


Creación exitosa

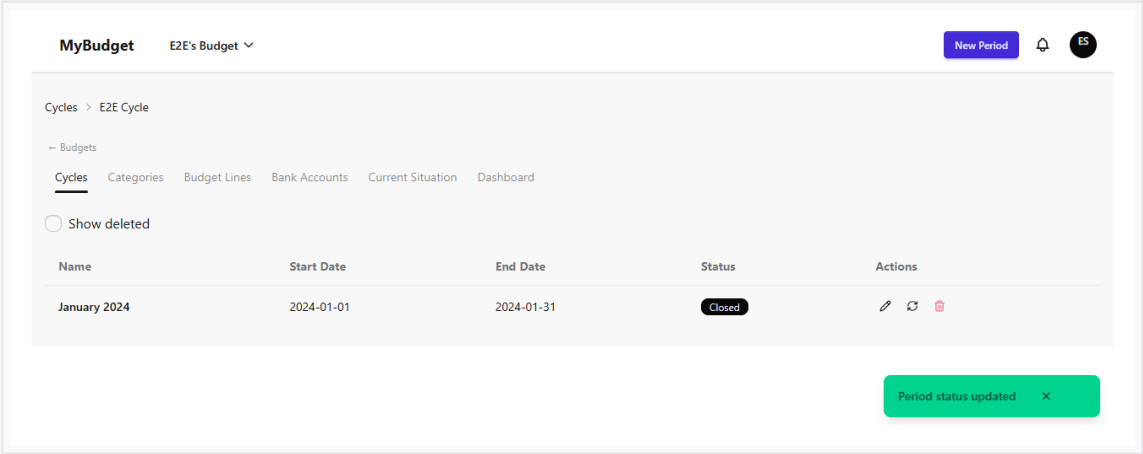


Error: nombre duplicado

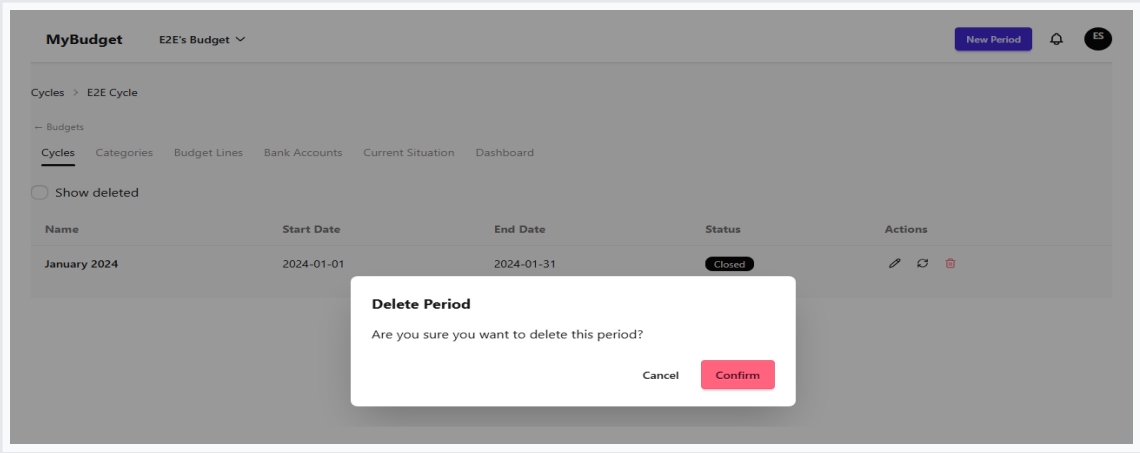
Períodos — estado y eliminación



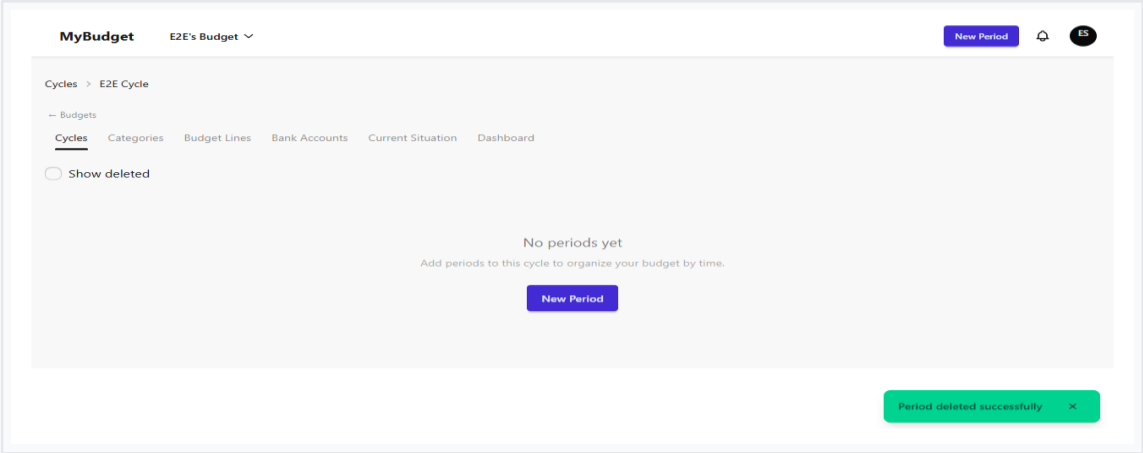
Cambio de estado



Estado actualizado

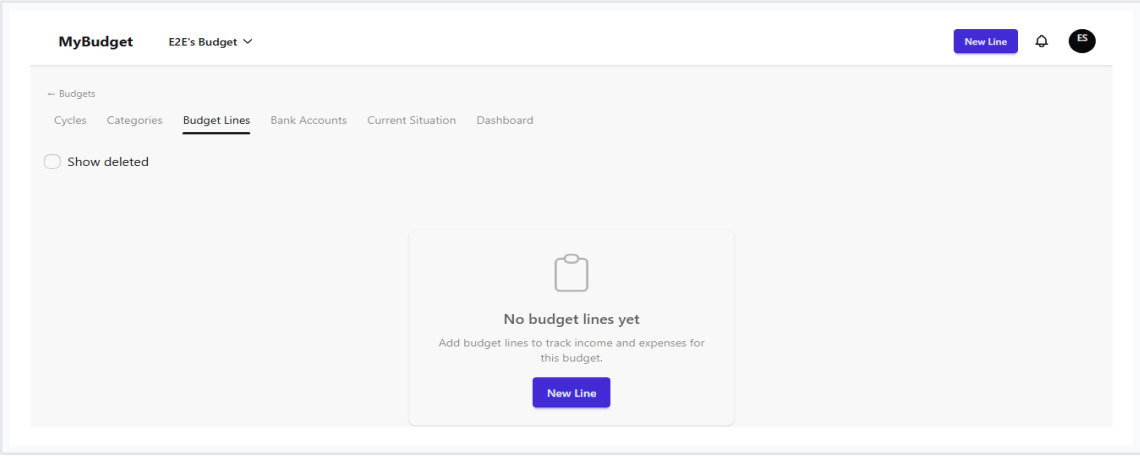


Confirmación de borrado

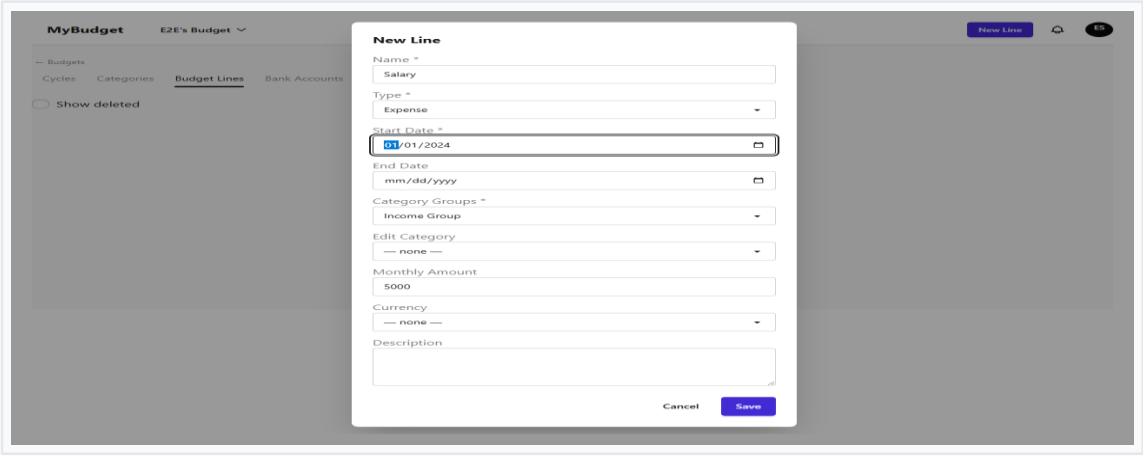


Borrado exitoso

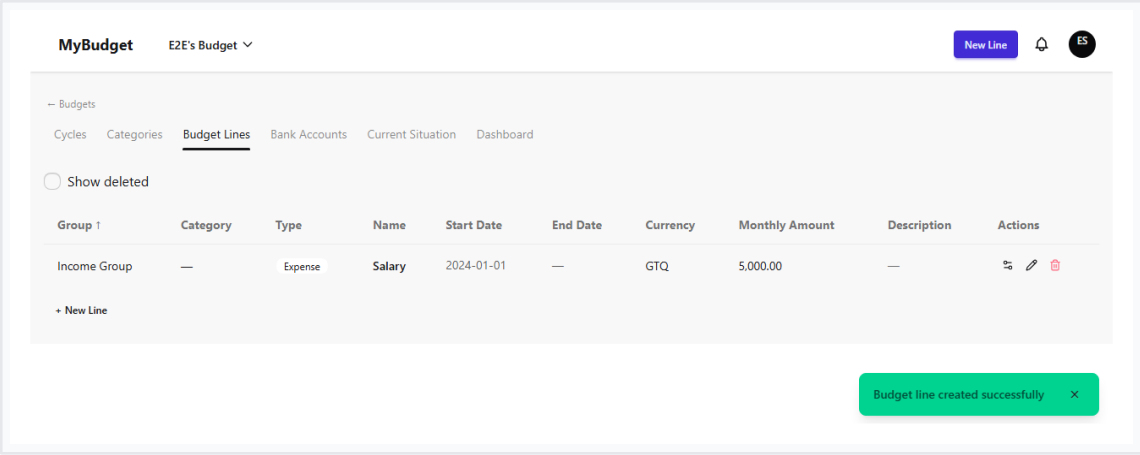
Rubros — creación



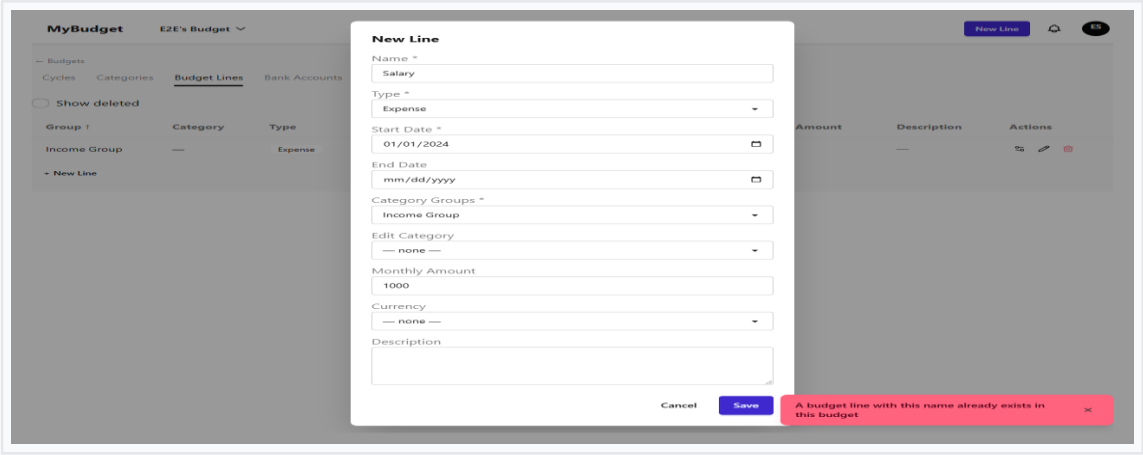
Lista vacía



Formulario de creación



Creación exitosa



Error: nombre duplicado

Rubros — edición y eliminación

MyBudget

E2E's Budget

New Line

ES

Budgets

Cycles

Categories

Budget Lines

Bank Accounts

Current Situation

Dashboard

Show deleted

Group 1	Category	Type	Name	Start Date	End Date	Currency	Monthly Amount	Description	Actions
Income Group		Expense	Monthly Salary	2024-01-01	—	GTQ	5,000.00	Description	✓ X

+ New Line

Edición en línea

MyBudget

E2E's Budget

New Line

ES

Budgets

Cycles

Categories

Budget Lines

Bank Accounts

Current Situation

Dashboard

Show deleted

Group 1	Category	Type	Name	Start Date	End Date	Currency	Monthly Amount	Description	Actions
Income Group	—	Expense	Monthly Salary	2024-01-01	—	GTQ	5,000.00	—	

+ New Line

Budget line updated successfully

Edición exitosa

MyBudget

E2E's Budget

New Line

ES

Budgets

Cycles

Categories

Budget Lines

Bank Accounts

Current Situation

Dashboard

Show deleted

Group 1	Category	Type	Name	Start Date	End Date	Currency	Monthly Amount	Description	Actions
Income Group	—	Expense	Monthly Salary	2024-01-01	—	GTQ	5,000.00	—	

+ New Line

Delete Line

Are you sure you want to delete this budget line?

Cancel

Confirm

Confirmación de borrado

MyBudget

E2E's Budget

New Line

ES

Budgets

Cycles

Categories

Budget Lines

Bank Accounts

Current Situation

Dashboard

Show deleted

Group 1	Category	Type	Name	Start Date	End Date	Currency	Monthly Amount	Description	Actions
Income Group	—	Expense	Monthly Salary	2024-01-01	—	GTQ	5,000.00	—	Restore

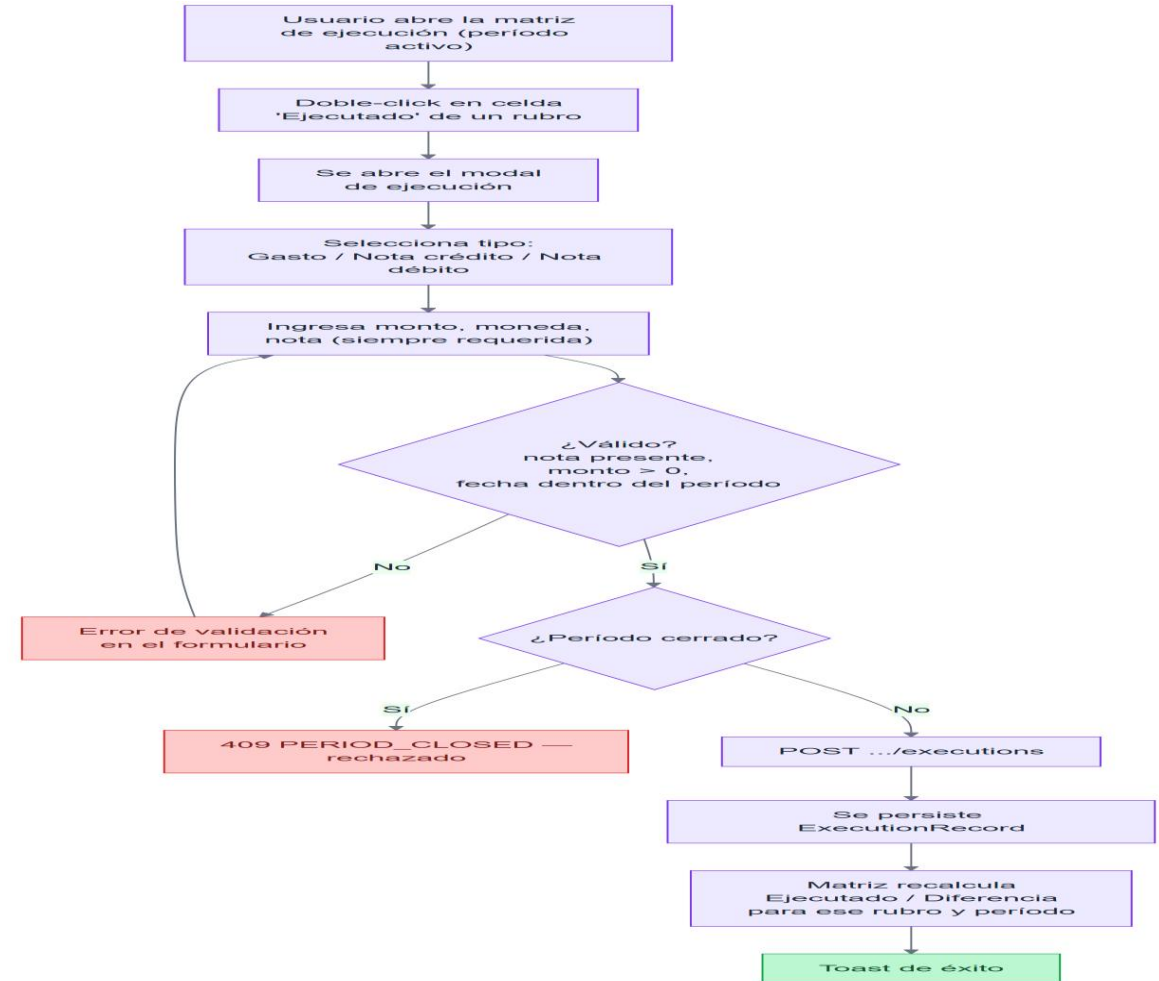
+ New Line

Budget line deleted successfully

Borrado exitoso

Ejecución (gasto real)

- Matriz multi-período con CRUD en línea sobre cada rubro.
- Notas de crédito y débito, además de gastos regulares.
- Tipo de cambio por entrada — cada ejecución guarda su propia tasa.
- Toggle de moneda para ver la matriz completa en GTQ o USD.



Matriz — vista y apertura

MyBudgetE2E's Budget

BudgetsCyclesCategoriesBudget LinesBank AccountsCurrent SituationDashboard

E2E Matrix Cycle

Previous periodNext period

Category / Line	January			February			March		
	Budgeted	Executed	Difference	Budgeted	Executed	Difference	Budgeted	Executed	Difference
Housing	Q 5,000.00	Q 0.00	Q 5,000.00	Q 5,000.00	Q 0.00	Q 5,000.00	Q 5,000.00	Q 0.00	Q 5,000.00
Rent	Q 5,000.00	Q 0.00	Q 5,000.00	Q 5,000.00	Q 0.00	Q 5,000.00	Q 5,000.00	Q 0.00	Q 5,000.00
Rent Payment	Q 5,000.00	Q 0.00	Q 5,000.00	Q 5,000.00	Q 0.00	Q 5,000.00	Q 5,000.00	Q 0.00	Q 5,000.00
Food	Q 2,000.00	Q 0.00	Q 2,000.00	Q 2,000.00	Q 0.00	Q 2,000.00	Q 2,000.00	Q 0.00	Q 2,000.00
Groceries	Q 2,000.00	Q 0.00	Q 2,000.00	Q 2,000.00	Q 0.00	Q 2,000.00	Q 2,000.00	Q 0.00	Q 2,000.00
Weekly Groceries	Q 2,000.00	Q 0.00	Q 2,000.00	Q 2,000.00	Q 0.00	Q 2,000.00	Q 2,000.00	Q 0.00	Q 2,000.00
Expenses SubTotal	Q 7,000.00	Q 0.00	Q 7,000.00	Q 7,000.00	Q 0.00	Q 7,000.00	Q 7,000.00	Q 0.00	Q 7,000.00
Preventive Savings SubTotal	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00
Long-term Savings SubTotal	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00
Total	Q 7,000.00	Q 0.00	Q 7,000.00	Q 7,000.00	Q 0.00	Q 7,000.00	Q 7,000.00	Q 0.00	Q 7,000.00

Vista de la matriz

MyBudgetE2E's Budget

BudgetsCyclesCategoriesBudget LinesBank AccountsCurrent SituationDashboard

E2E Matrix Cycle

Previous periodNext period

Category / Line	January			February			March		
	Budgeted	Executed	Difference	Budgeted	Executed	Difference	Budgeted	Executed	Difference
Housing	Q 5,000.00	Q 0.00	Q 5,000.00	Q 5,000.00	Q 0.00	Q 5,000.00	Q 5,000.00	Q 0.00	Q 5,000.00
Rent	Q 5,000.00	Q 0.00	Q 5,000.00	Q 5,000.00	Q 0.00	Q 5,000.00	Q 5,000.00	Q 0.00	Q 5,000.00
Rent Payment	Q 5,000.00	Q 0.00	Q 5,000.00	Q 5,000.00	Q 0.00	Q 5,000.00	Q 5,000.00	Q 0.00	Q 5,000.00
Food	Q 2,000.00	Q 0.00	Q 2,000.00	Q 2,000.00	Q 0.00	Q 2,000.00	Q 2,000.00	Q 0.00	Q 2,000.00
Groceries	Q 2,000.00	Q 0.00	Q 2,000.00	Q 2,000.00	Q 0.00	Q 2,000.00	Q 2,000.00	Q 0.00	Q 2,000.00
Weekly Groceries	Q 2,000.00	Q 0.00	Q 2,000.00	Q 2,000.00	Q 0.00	Q 2,000.00	Q 2,000.00	Q 0.00	Q 2,000.00
Expenses SubTotal	Q 7,000.00	Q 0.00	Q 7,000.00	Q 7,000.00	Q 0.00	Q 7,000.00	Q 7,000.00	Q 0.00	Q 7,000.00
Preventive Savings SubTotal	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00
Long-term Savings SubTotal	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00
Total	Q 7,000.00	Q 0.00	Q 7,000.00	Q 7,000.00	Q 0.00	Q 7,000.00	Q 7,000.00	Q 0.00	Q 7,000.00

Executions

Include deleted

No entries yet

Add entry

Operation date08/09/2026

Entry type *Expense

CurrencyGTQ — Q

Amount *

Note *

Save

Modal de ejecución

Matriz — validación

MyBudgetE2E's Budget

Budgets

CyclesCategoriesBudget LinesBank AccountsCurrent SituationDashboard

E2E Matrix Cycle

Previous periodNext period

Category / Line

Housing

Rent

Rent Payment

Food

Groceries

Weekly Groceries

Expenses SubTotal

Preventive Savings SubTotal

Long-term Savings SubTotal

Total

March

Budgeted	Executed	Difference
Q 5,000.00	Q 0.00	Q 5,000.00
Q 5,000.00	Q 0.00	Q 5,000.00
Q 5,000.00	Q 0.00	Q 5,000.00
Q 2,000.00	Q 0.00	Q 2,000.00
Q 2,000.00	Q 0.00	Q 2,000.00
Q 2,000.00	Q 0.00	Q 2,000.00
Q 7,000.00	Q 0.00	Q 7,000.00
Q 0.00	Q 0.00	Q 0.00
Q 0.00	Q 0.00	Q 0.00
Q 7,000.00	Q 0.00	Q 7,000.00

Executions

Include deleted

No entries yet

Add entry

Operation date

08/09/2026

Entry type *

Credit Note

Currency

GTQ — Q

Amount *

50

Note *

Note is required

Save

Error de validación

MyBudgetE2E's Budget

Budgets

CyclesCategoriesBudget LinesBank AccountsCurrent SituationDashboard

E2E Matrix Cycle

Previous periodNext period

Category / Line

Housing

Rent

Rent Payment

Food

Groceries

Weekly Groceries

Expenses SubTotal

Preventive Savings SubTotal

Long-term Savings SubTotal

Total

March

Budgeted	Executed	Difference
Q 5,000.00	Q 0.00	Q 5,000.00
Q 5,000.00	Q 0.00	Q 5,000.00
Q 5,000.00	Q 0.00	Q 5,000.00
Q 2,000.00	Q 0.00	Q 2,000.00
Q 2,000.00	Q 0.00	Q 2,000.00
Q 2,000.00	Q 0.00	Q 2,000.00
Q 7,000.00	Q 0.00	Q 7,000.00
Q 0.00	Q 0.00	Q 0.00
Q 0.00	Q 0.00	Q 0.00
Q 7,000.00	Q 0.00	Q 7,000.00

Executions

Include deleted

No entries yet

Add entry

Operation date

08/09/2026

Entry type *

Expense

Currency

GTQ — Q

Amount *

100

Note *

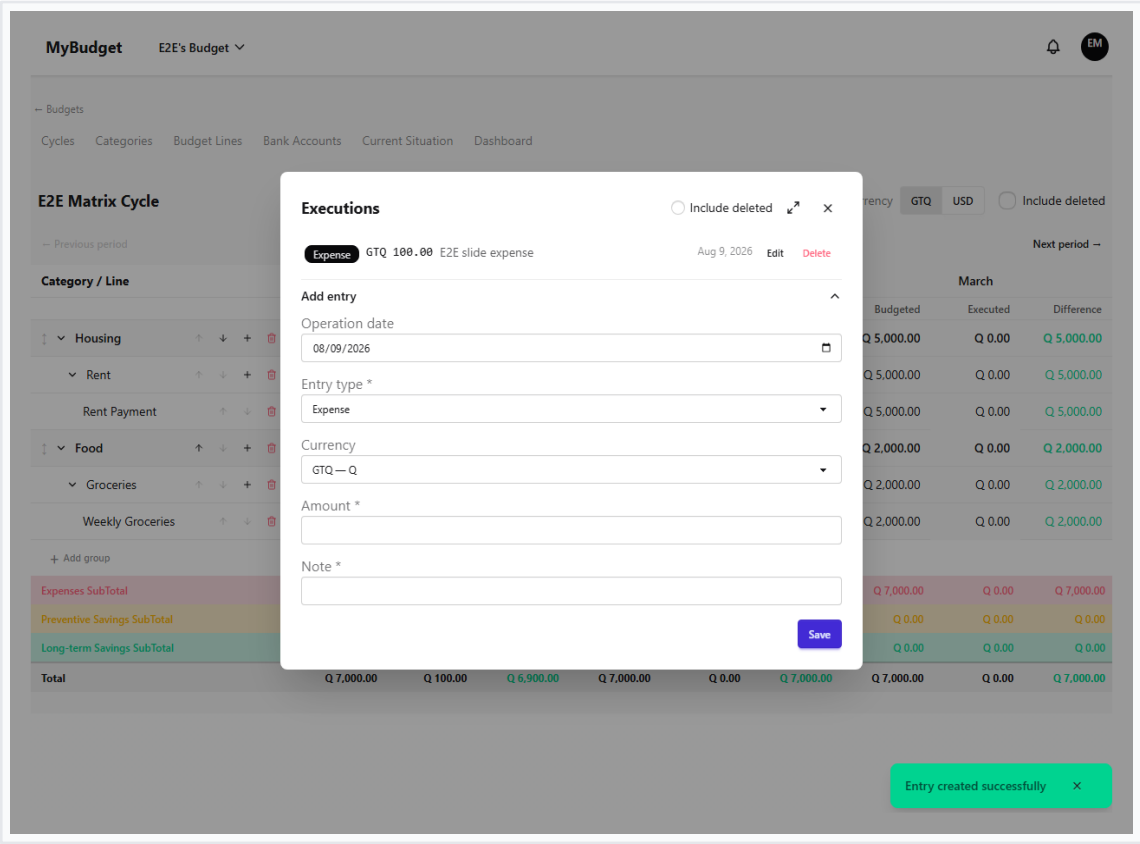
E2E slide expense

Note is required

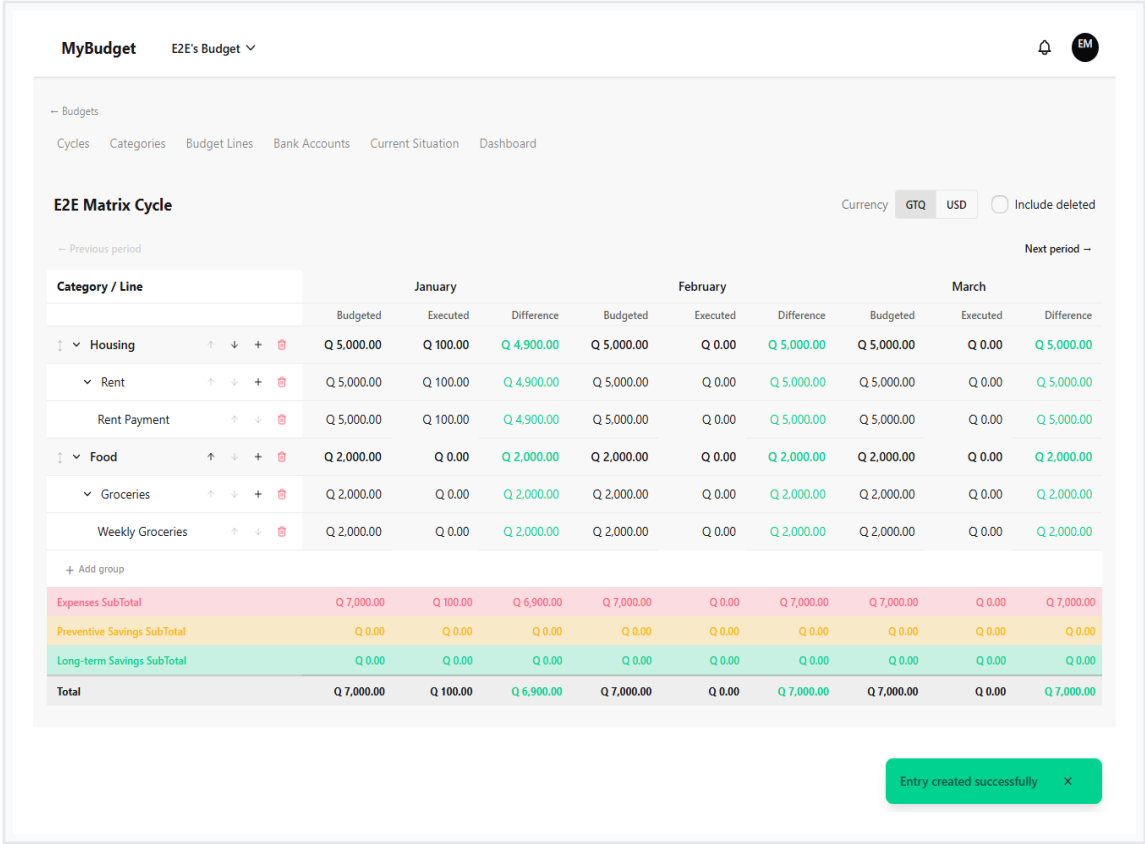
Save

Formulario completado

Matriz — creación exitosa



Creación exitosa



Matriz actualizada

Matriz — organización

MyBudget

E2E's Budget

EM

Budgets

CyclesCategoriesBudget LinesBank AccountsCurrent SituationDashboard

E2E Matrix Cycle

Currency

GTQ

USD

☐ Include deleted

Previous period

Next period

Category / Line	January			February			March		
	Budgeted	Executed	Difference	Budgeted	Executed	Difference	Budgeted	Executed	Difference
> Housing	Q 5,000.00	Q 100.00	Q 4,900.00	Q 5,000.00	Q 0.00	Q 5,000.00	Q 5,000.00	Q 0.00	Q 5,000.00
Food	Q 2,000.00	Q 0.00	Q 2,000.00	Q 2,000.00	Q 0.00	Q 2,000.00	Q 2,000.00	Q 0.00	Q 2,000.00
Groceries	Q 2,000.00	Q 0.00	Q 2,000.00	Q 2,000.00	Q 0.00	Q 2,000.00	Q 2,000.00	Q 0.00	Q 2,000.00
Weekly Groceries	Q 2,000.00	Q 0.00	Q 2,000.00	Q 2,000.00	Q 0.00	Q 2,000.00	Q 2,000.00	Q 0.00	Q 2,000.00
+ Add group									
Expenses SubTotal	Q 7,000.00	Q 100.00	Q 6,900.00	Q 7,000.00	Q 0.00	Q 7,000.00	Q 7,000.00	Q 0.00	Q 7,000.00
Preventive Savings SubTotal	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00
Long-term Savings SubTotal	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00
Total	Q 7,000.00	Q 100.00	Q 6,900.00	Q 7,000.00	Q 0.00	Q 7,000.00	Q 7,000.00	Q 0.00	Q 7,000.00

Entry create

Grupo colapsado

Matriz — moneda

MyBudget

E2E's Budget

EM

← Budgets

CyclesCategoriesBudget LinesBank AccountsCurrent SituationDashboard

E2E Matrix Cycle

CurrencyGTQUSD7.5GTQ = 1 USDInclude deleted

← Previous periodNext period →

Category / Line	January			February			March		
	Budgeted	Executed	Difference	Budgeted	Executed	Difference	Budgeted	Executed	Difference
⌵ Housing	\$ 666.67	\$ 13.33	\$ 653.33	\$ 666.67	\$ 0.00	\$ 666.67	\$ 666.67	\$ 0.00	\$ 666.67
⌵ Rent	\$ 666.67	\$ 13.33	\$ 653.33	\$ 666.67	\$ 0.00	\$ 666.67	\$ 666.67	\$ 0.00	\$ 666.67
Rent Payment	\$ 666.67	\$ 13.33	\$ 653.33	\$ 666.67	\$ 0.00	\$ 666.67	\$ 666.67	\$ 0.00	\$ 666.67
⌵ Food	\$ 266.67	\$ 0.00	\$ 266.67	\$ 266.67	\$ 0.00	\$ 266.67	\$ 266.67	\$ 0.00	\$ 266.67
⌵ Groceries	\$ 266.67	\$ 0.00	\$ 266.67	\$ 266.67	\$ 0.00	\$ 266.67	\$ 266.67	\$ 0.00	\$ 266.67
Weekly Groceries	\$ 266.67	\$ 0.00	\$ 266.67	\$ 266.67	\$ 0.00	\$ 266.67	\$ 266.67	\$ 0.00	\$ 266.67
+ Add group									
Expenses SubTotal	\$ 933.33	\$ 13.33	\$ 920.00	\$ 933.33	\$ 0.00	\$ 933.33	\$ 933.33	\$ 0.00	\$ 933.33
Preventive Savings SubTotal	\$ 0.00	\$ 0.00	\$ 0.00	\$ 0.00	\$ 0.00	\$ 0.00	\$ 0.00	\$ 0.00	\$ 0.00
Long-term Savings SubTotal	\$ 0.00	\$ 0.00	\$ 0.00	\$ 0.00	\$ 0.00	\$ 0.00	\$ 0.00	\$ 0.00	\$ 0.00
Total	\$ 933.33	\$ 13.33	\$ 920.00	\$ 933.33	\$ 0.00	\$ 933.33	\$ 933.33	\$ 0.00	\$ 933.33

Entry created successfully

Vista en USD

MyBudget

E2E's Budget

EM

← Budgets

CyclesCategoriesBudget LinesBank AccountsCurrent SituationDashboard

E2E Matrix Cycle

CurrencyGTQUSDInclude deleted

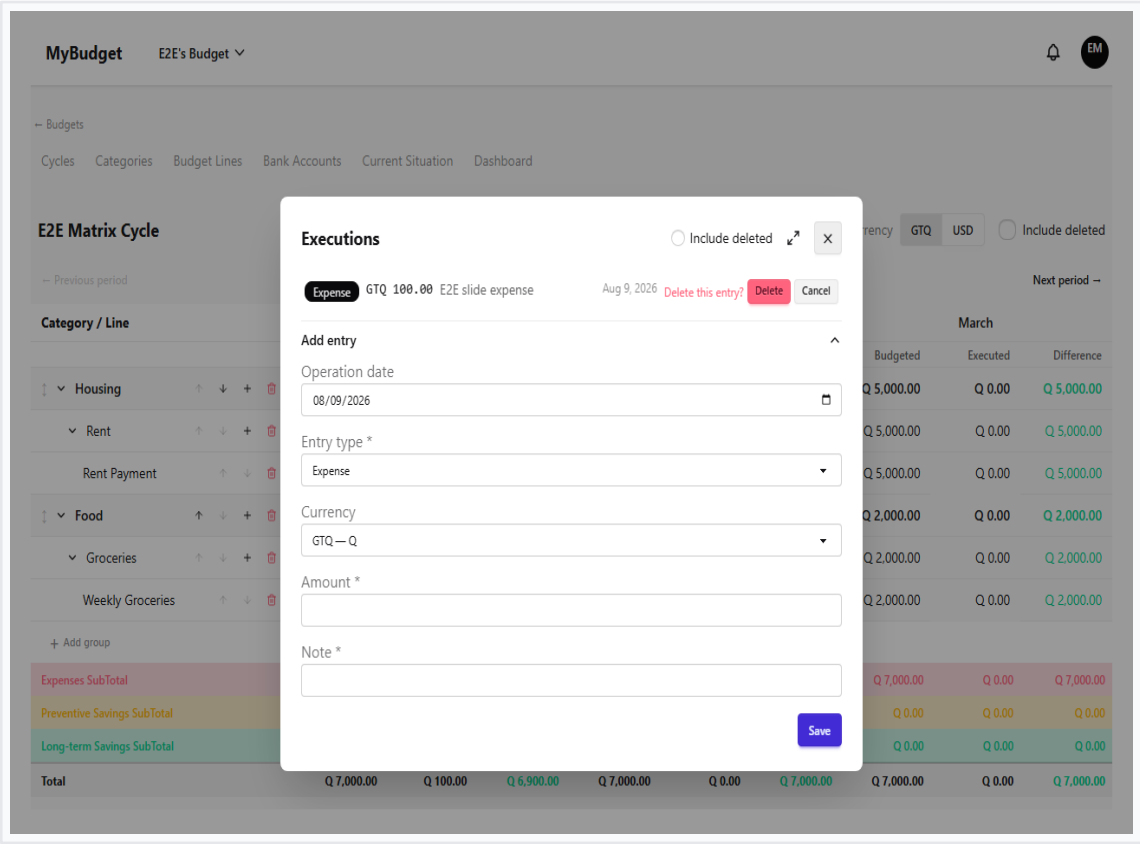
← Previous periodNext period →

Category / Line	January			February			March		
	Budgeted	Executed	Difference	Budgeted	Executed	Difference	Budgeted	Executed	Difference
⌵ Housing	Q 5,000.00	Q 100.00	Q 4,900.00	Q 5,000.00	Q 0.00	Q 5,000.00	Q 5,000.00	Q 0.00	Q 5,000.00
⌵ Rent	Q 5,000.00	Q 100.00	Q 4,900.00	Q 5,000.00	Q 0.00	Q 5,000.00	Q 5,000.00	Q 0.00	Q 5,000.00
Rent Payment	Q 5,000.00	Q 100.00	Q 4,900.00	Q 5,000.00	Q 0.00	Q 5,000.00	Q 5,000.00	Q 0.00	Q 5,000.00
⌵ Food	Q 2,000.00	Q 0.00	Q 2,000.00	Q 2,000.00	Q 0.00	Q 2,000.00	Q 2,000.00	Q 0.00	Q 2,000.00
⌵ Groceries	Q 2,000.00	Q 0.00	Q 2,000.00	Q 2,000.00	Q 0.00	Q 2,000.00	Q 2,000.00	Q 0.00	Q 2,000.00
Weekly Groceries	Q 2,000.00	Q 0.00	Q 2,000.00	Q 2,000.00	Q 0.00	Q 2,000.00	Q 2,000.00	Q 0.00	Q 2,000.00
+ Add group									
Expenses SubTotal	Q 7,000.00	Q 100.00	Q 6,900.00	Q 7,000.00	Q 0.00	Q 7,000.00	Q 7,000.00	Q 0.00	Q 7,000.00
Preventive Savings SubTotal	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00
Long-term Savings SubTotal	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00	Q 0.00
Total	Q 7,000.00	Q 100.00	Q 6,900.00	Q 7,000.00	Q 0.00	Q 7,000.00	Q 7,000.00	Q 0.00	Q 7,000.00

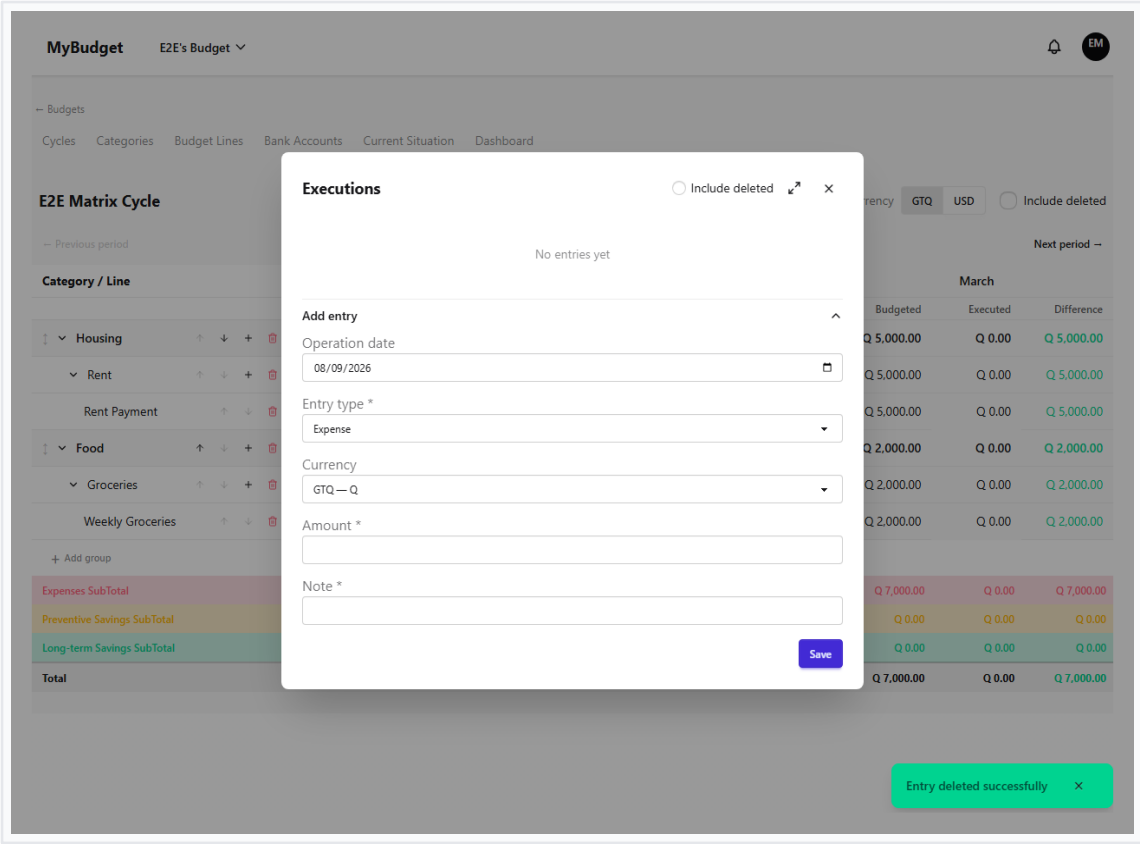
Entry created successfully

Vista en GTQ

Matriz — eliminación



Confirmación de borrado



Borrado exitoso

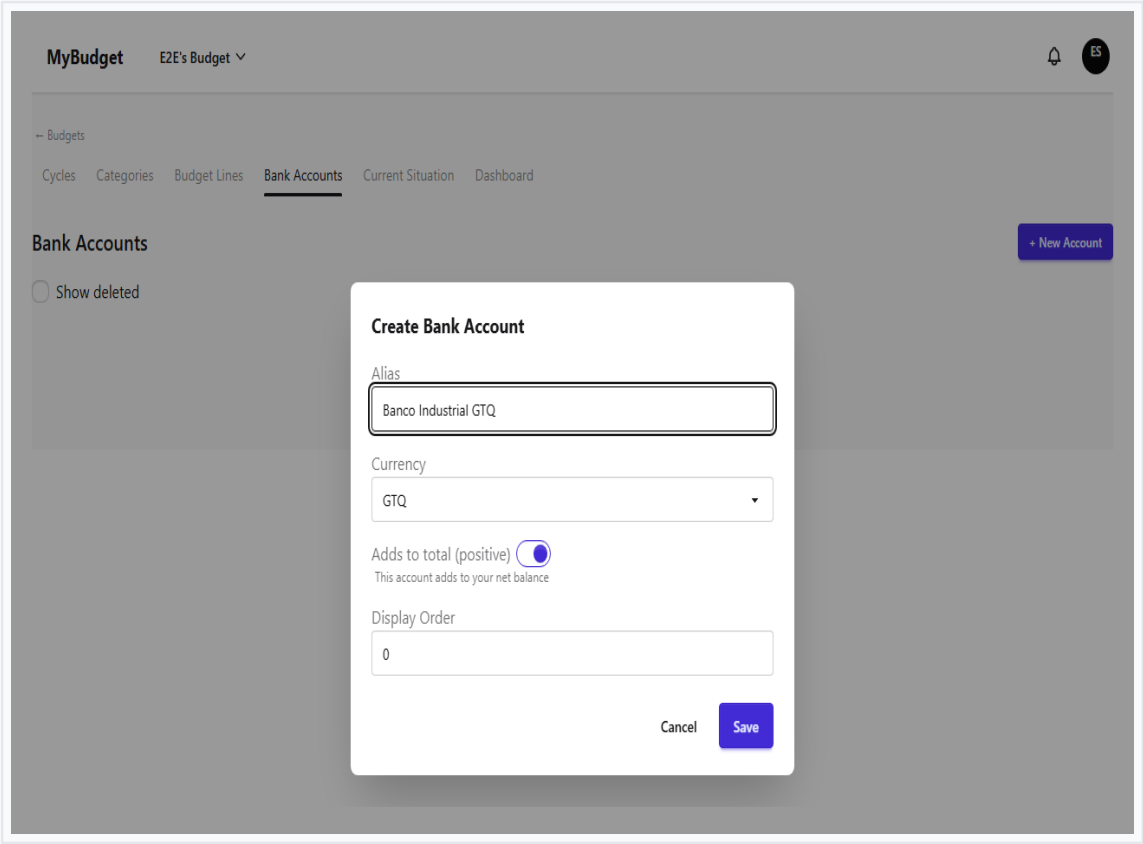
Situación actual — el "corte"

- Catálogo de cuentas bancarias por presupuesto.
- Snapshot diario de saldos vs. presupuestado/ejecutado — el "corte".
- Soporte multi-moneda con tipo de cambio por corte.
- Los totales se persisten al momento de guardar — no se recalculan retroactivamente si cambian ejecuciones posteriores.

Cuentas bancarias — creación

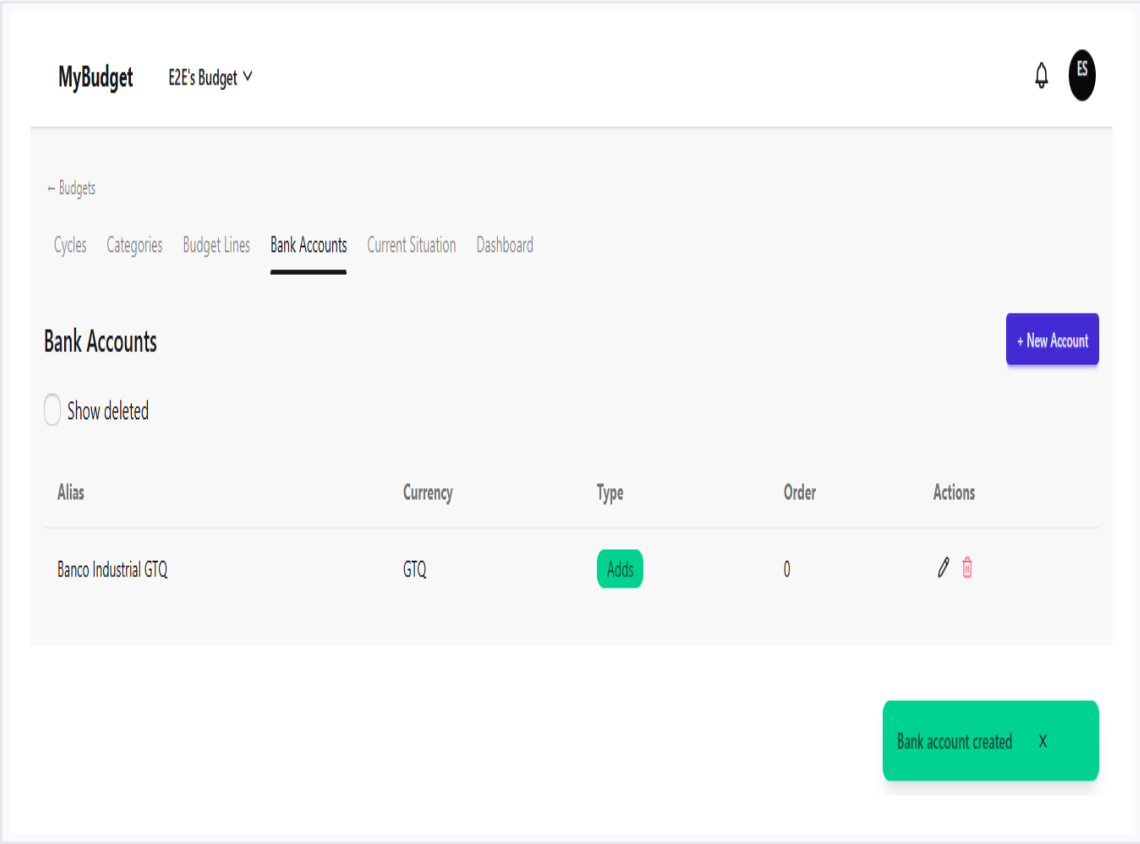


Lista vacía

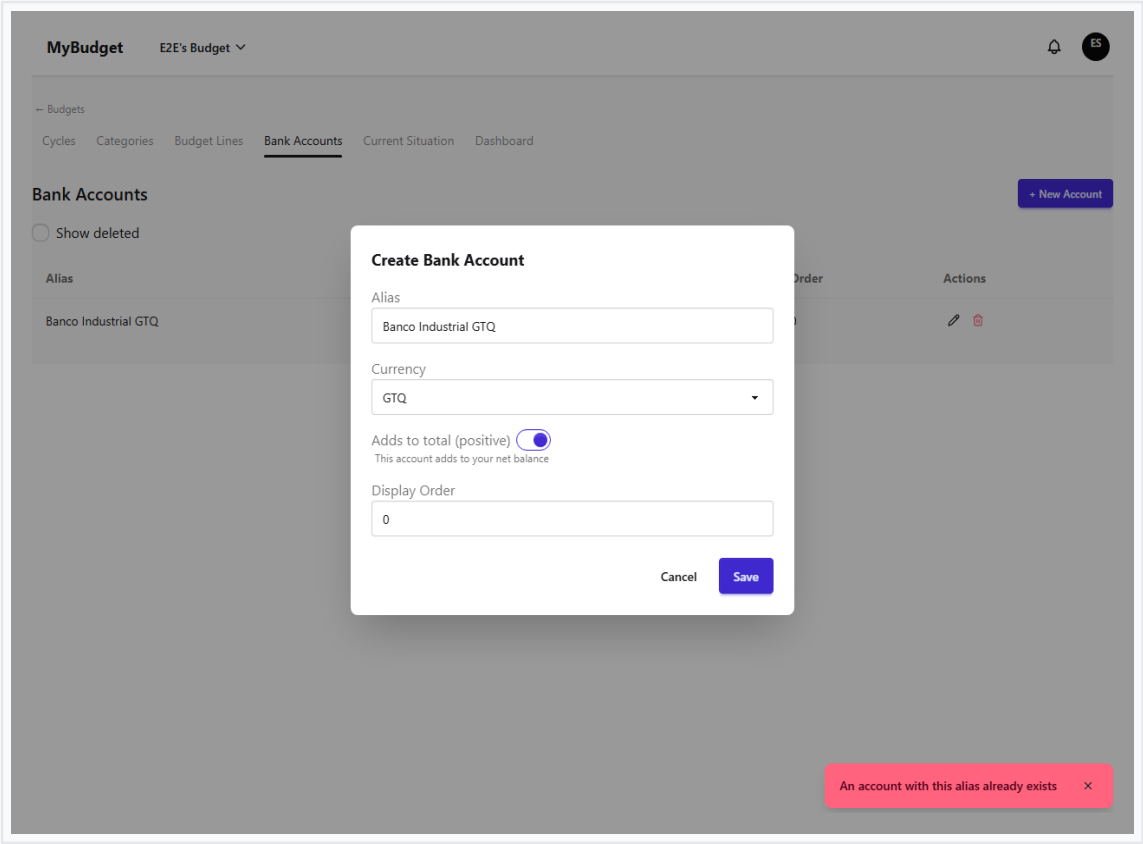


Formulario de creación

Cuentas bancarias — resultado



Creación exitosa



Error: alias duplicado

Cuentas bancarias — mostrar eliminadas

MyBudget

E2E's Budget

← Budgets

Cycles

Categories

Budget Lines

Bank Accounts

Current Situation

Dashboard

Bank Accounts

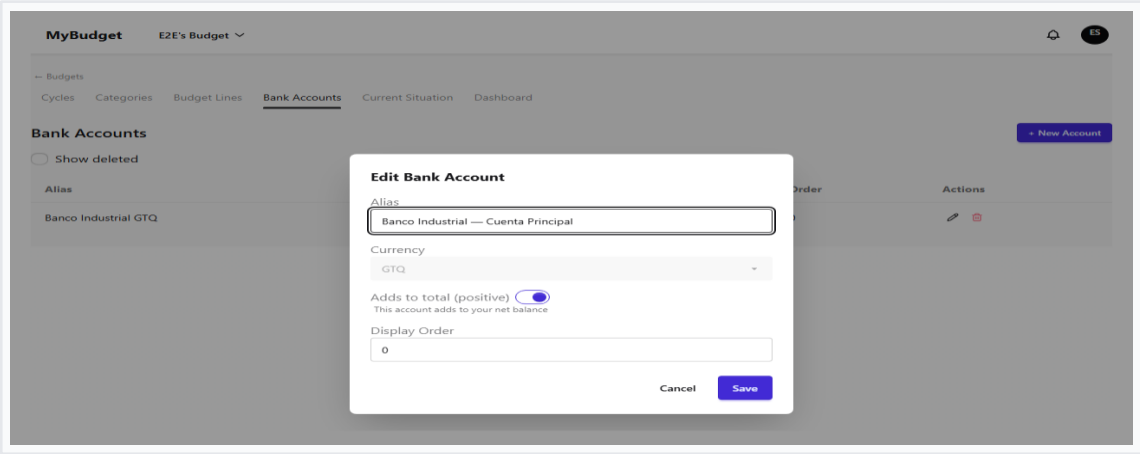
+ New Account

☒ Show deleted

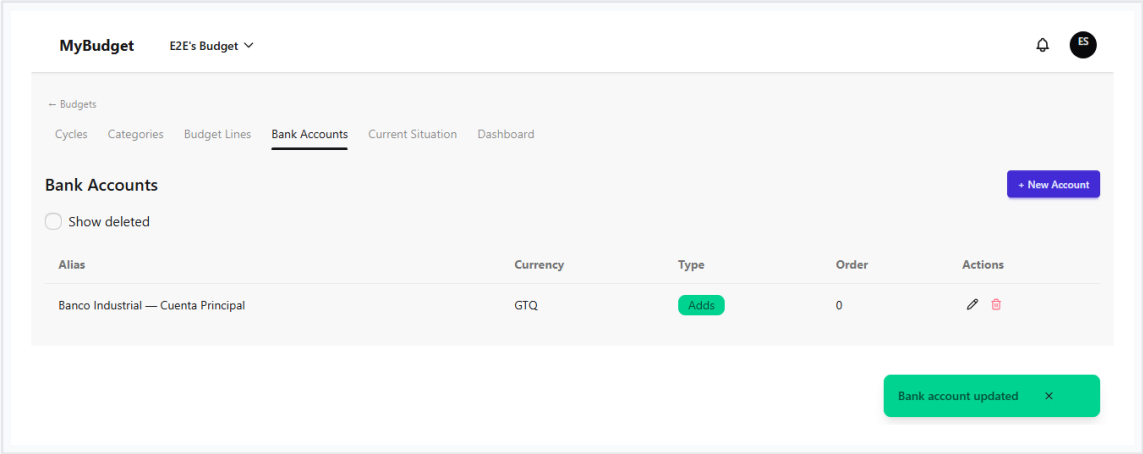
Alias	Currency	Type	Order	Actions
Banco Industrial — Cuenta Principal	GTQ	Added	0	Restore

Mostrar eliminadas

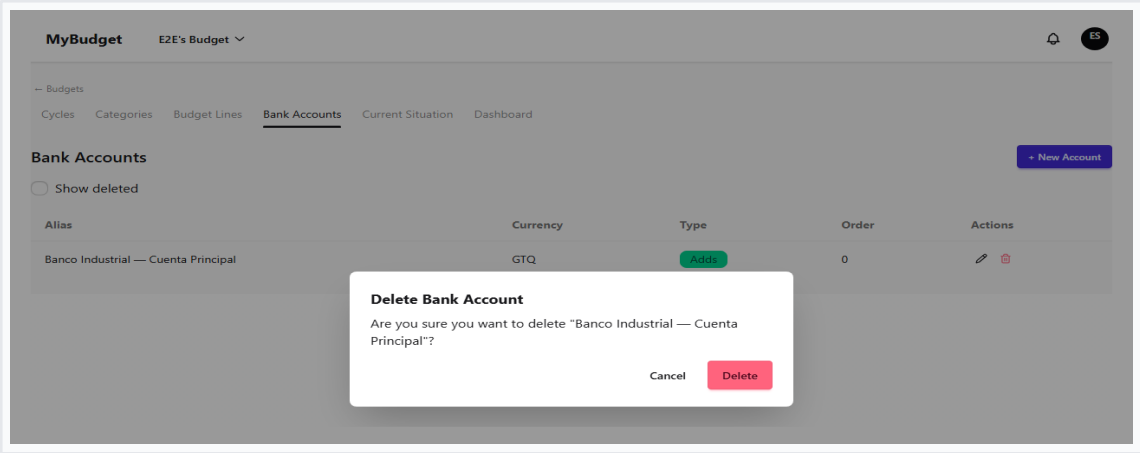
Cuentas bancarias — edición y ciclo de vida



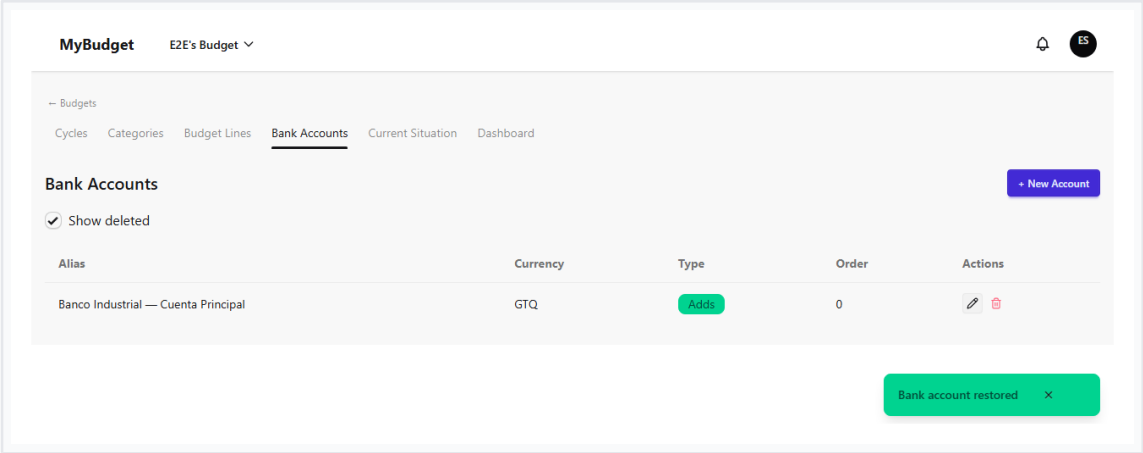
Formulario de edición



Edición exitosa



Confirmación de borrado



Restauración exitosa

Corte — borrador y formulario

Borrador

Formulario completado

SITUACIÓN ACTUAL

Corte — guardado

MyBudget

E2E's Budget

ES

Budgets

Cycles

Categories

Budget Lines

Bank Accounts

Current Situation

Dashboard

Current Situation

Draft

Cut Date

08/09/2026

Exchange Rate

7.8

Assets (add to total)

Caja GTQ

GTQ

1,500.50

CUT TOTALS

	Q	USD
Total Assets (sum)	1,500.50	192.37
Total Liabilities (debt)	- 0.00	- 0.00
Total Budgeted	0.00	0.00
Total Registered	0.00	0.00
Budget Commitment	- 0.00	- 0.00
Total Available	1,500.50	192.37
Total Debt	- 0.00	- 0.00
Total Net	1,500.50	192.37

Totals reflect account balances and budget execution as of this cut's last save. Later changes to accounts or execution records won't affect a saved cut until it is saved again.

No active budget period covers this cut date. Please check your cycle and period setup before saving.

Save

Error al guardar

MyBudget

E2E's Budget

ES

Budgets

Cycles

Categories

Budget Lines

Bank Accounts

Current Situation

Dashboard

Current Situation

2026-08-09

Delete Cut

Cut Date

08/09/2026

Exchange Rate

7.8

Assets (add to total)

Caja GTQ

GTQ

1,500.50

CUT TOTALS

	Q	USD
Total Assets (sum)	1,500.50	192.37
Total Liabilities (debt)	- 0.00	- 0.00
Total Budgeted	0.00	0.00
Total Registered	0.00	0.00
Budget Commitment	- 0.00	- 0.00
Total Available	1,500.50	192.37
Total Debt	- 0.00	- 0.00
Total Net	1,500.50	192.37

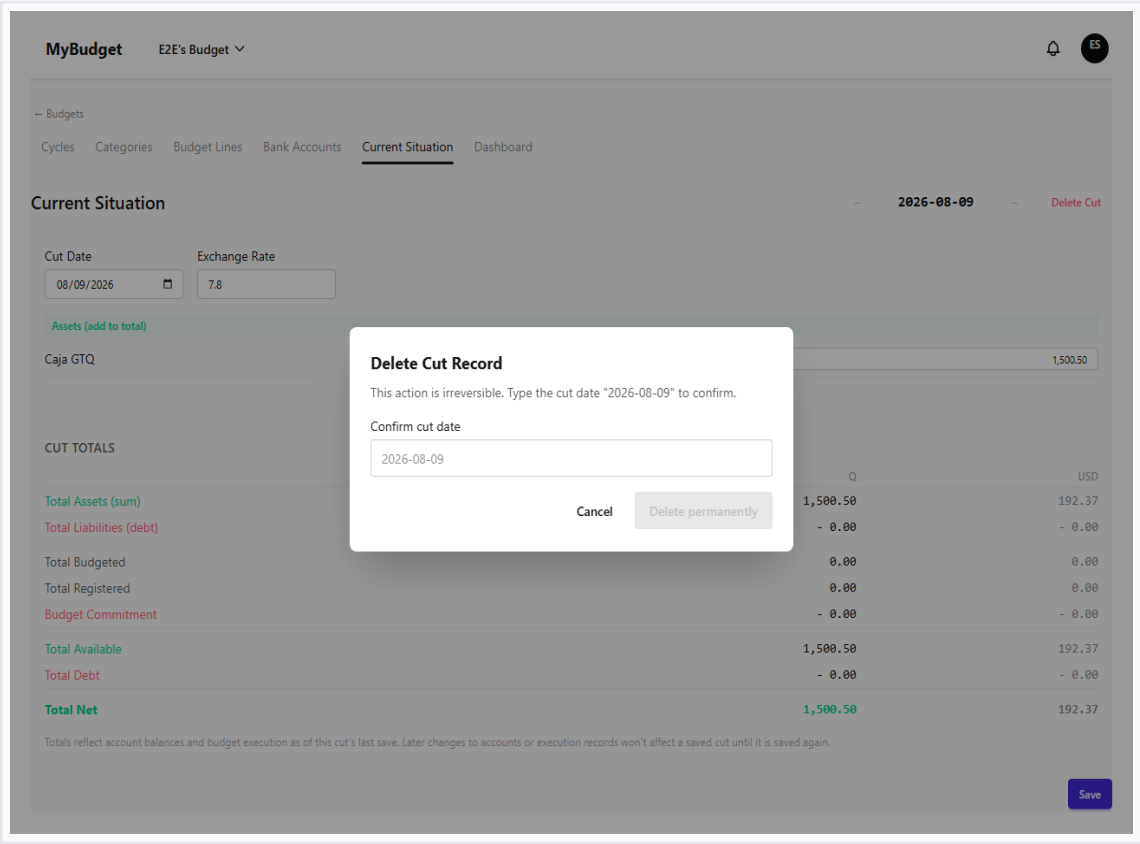
Totals reflect account balances and budget execution as of this cut's last save. Later changes to accounts or execution records won't affect a saved cut until it is saved again.

Save

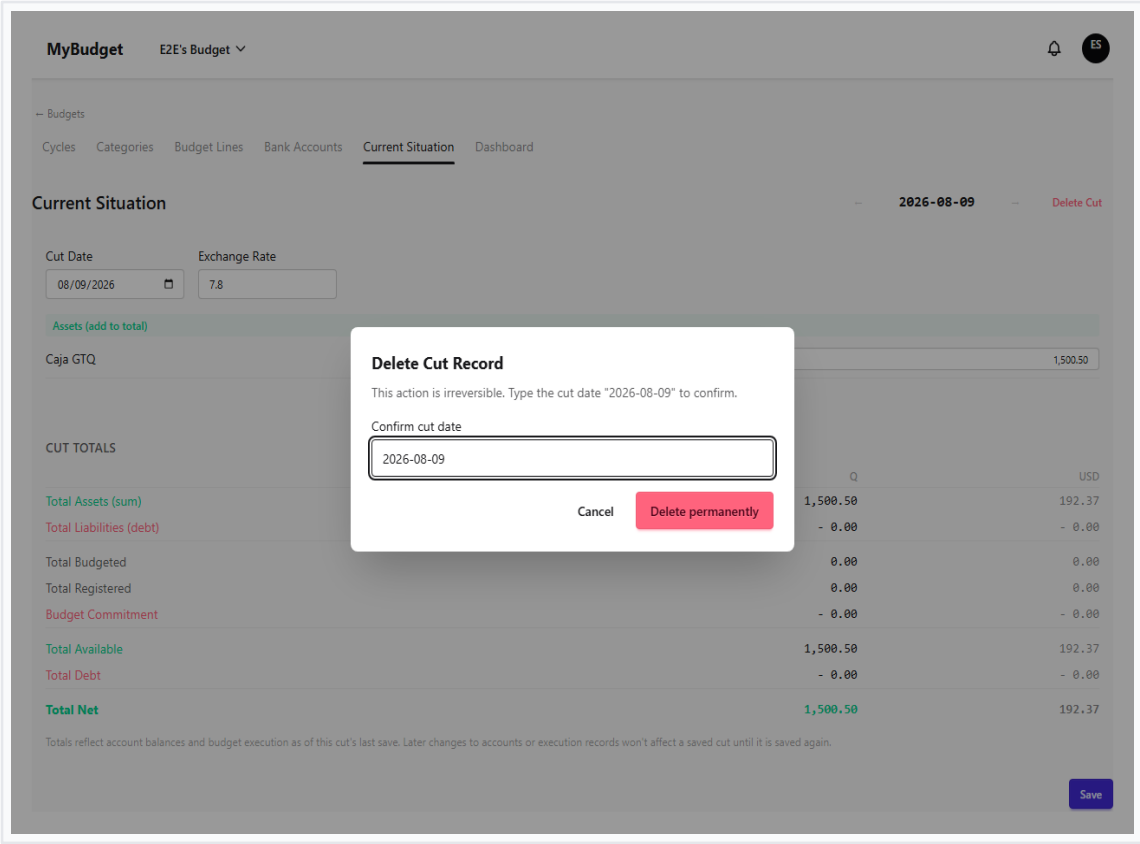
Cut record saved

Guardado exitoso

Corte — confirmar eliminación

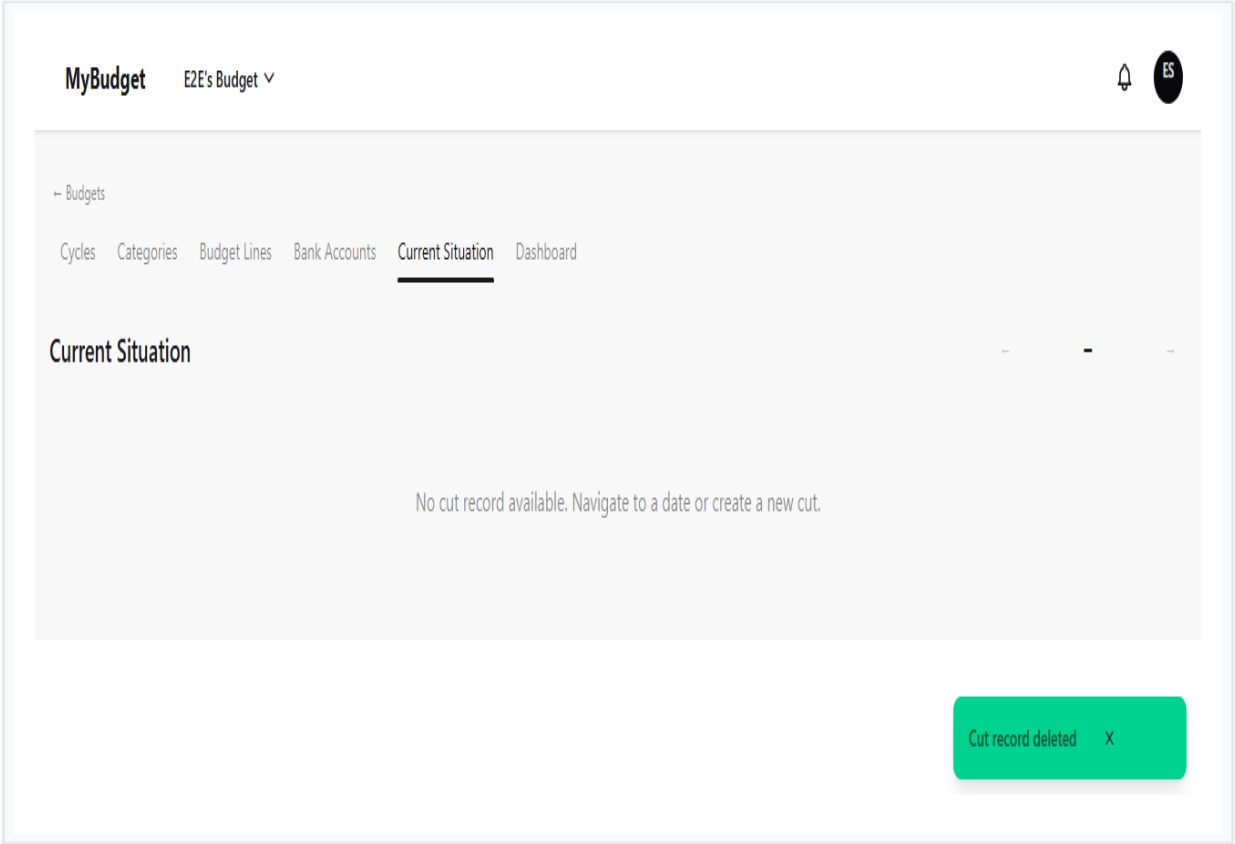


Confirmación (vacía)



Fecha escrita

Corte — eliminación exitosa



Borrado exitoso

Dashboard analítico

- Tendencia histórica de los totales del "corte" a lo largo del tiempo.
- Banda de comportamiento promedio (mínimo/máximo/promedio por período).
- Comparación de rubros por período y por ciclo, dentro y entre ciclos.
- Manejo explícito de tipo de cambio: congelado al corte vs. transaccional — nunca mezclados.

Tendencia histórica

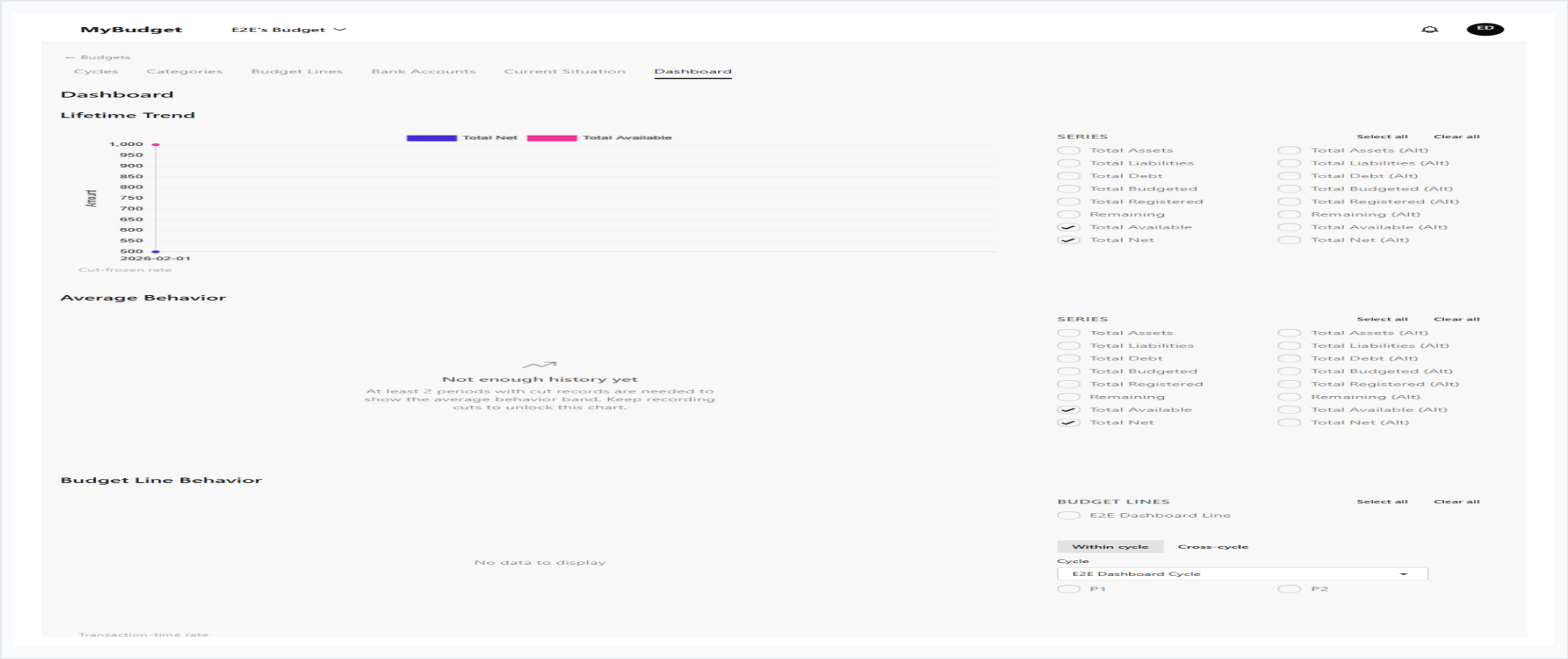


Selector de series — estado vacío



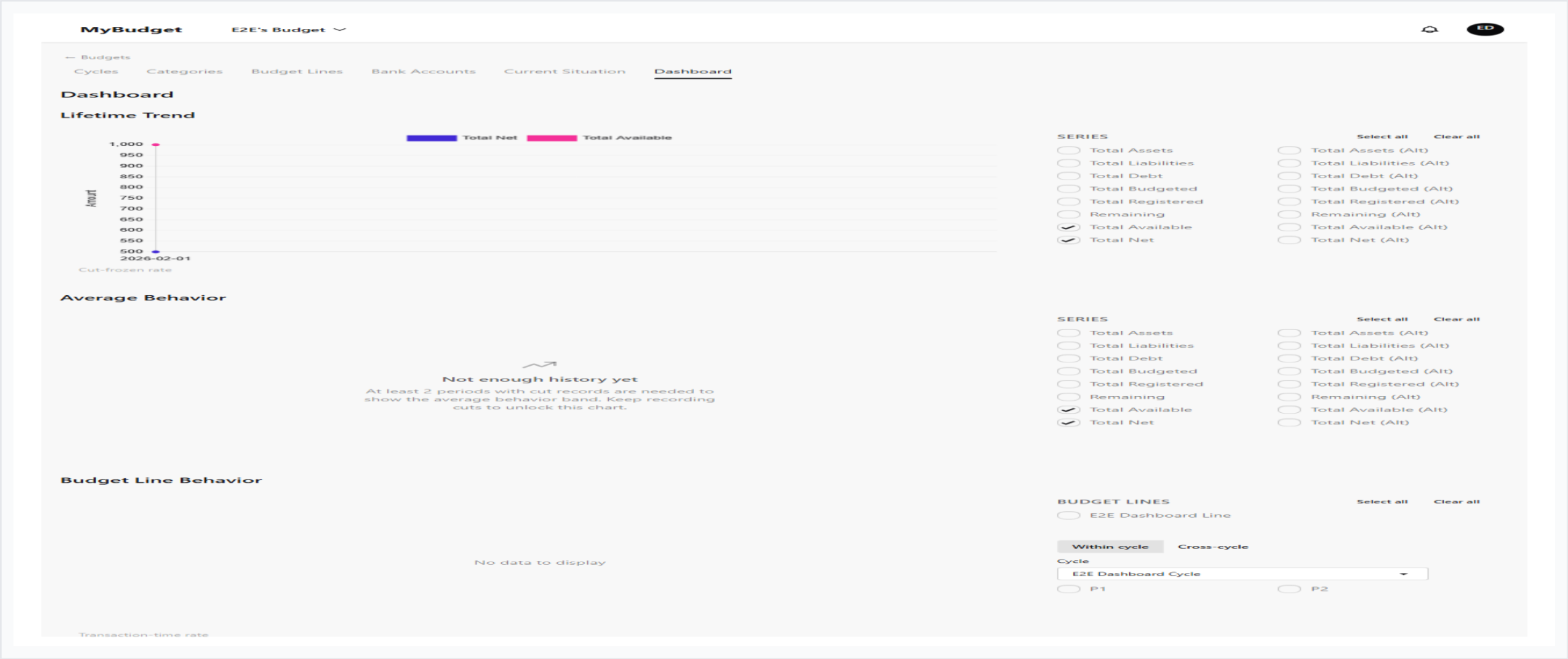
Al deselectionar todas las series, el gráfico muestra un estado vacío explícito — prueba de que el selector controla el gráfico.

Historial insuficiente

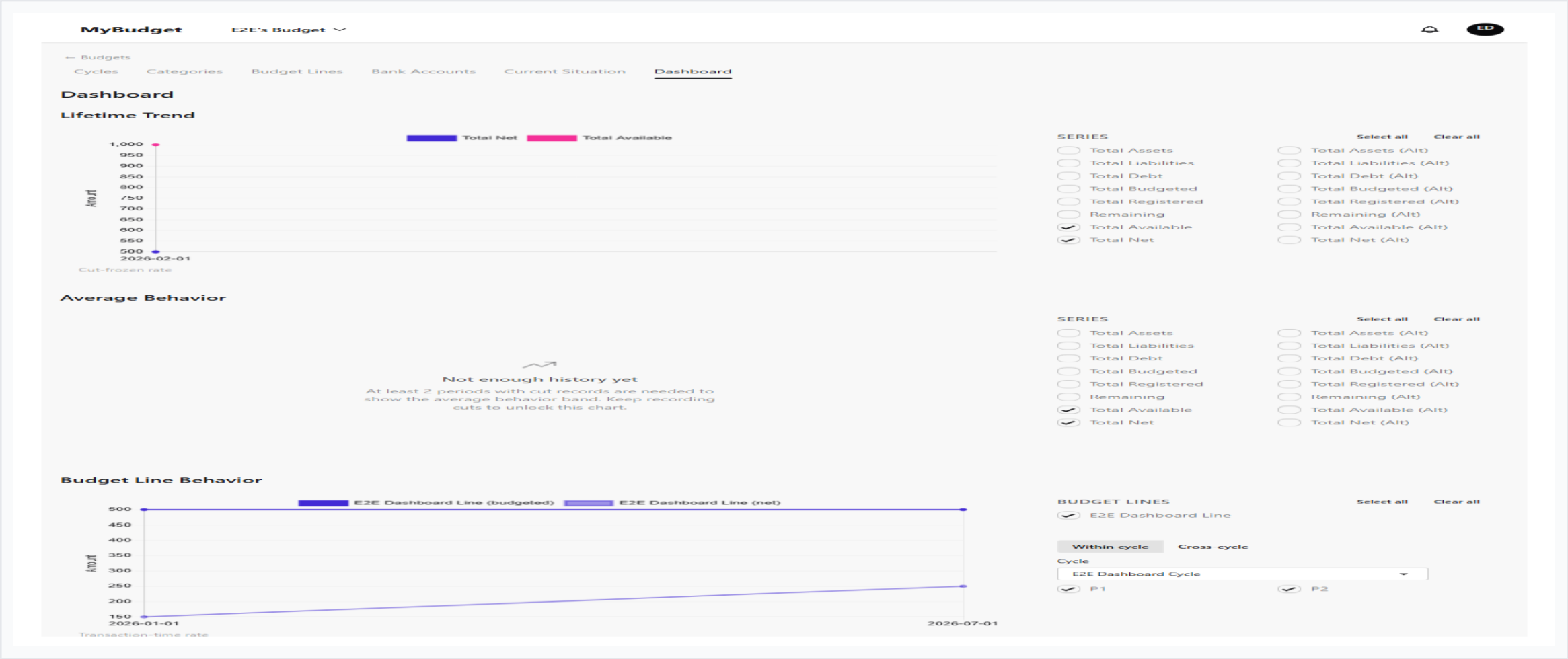


Con menos de 2 cortes, se muestra un estado vacío explícito en vez de una banda calculada engañosa.

Comparación de rubros — sin selección

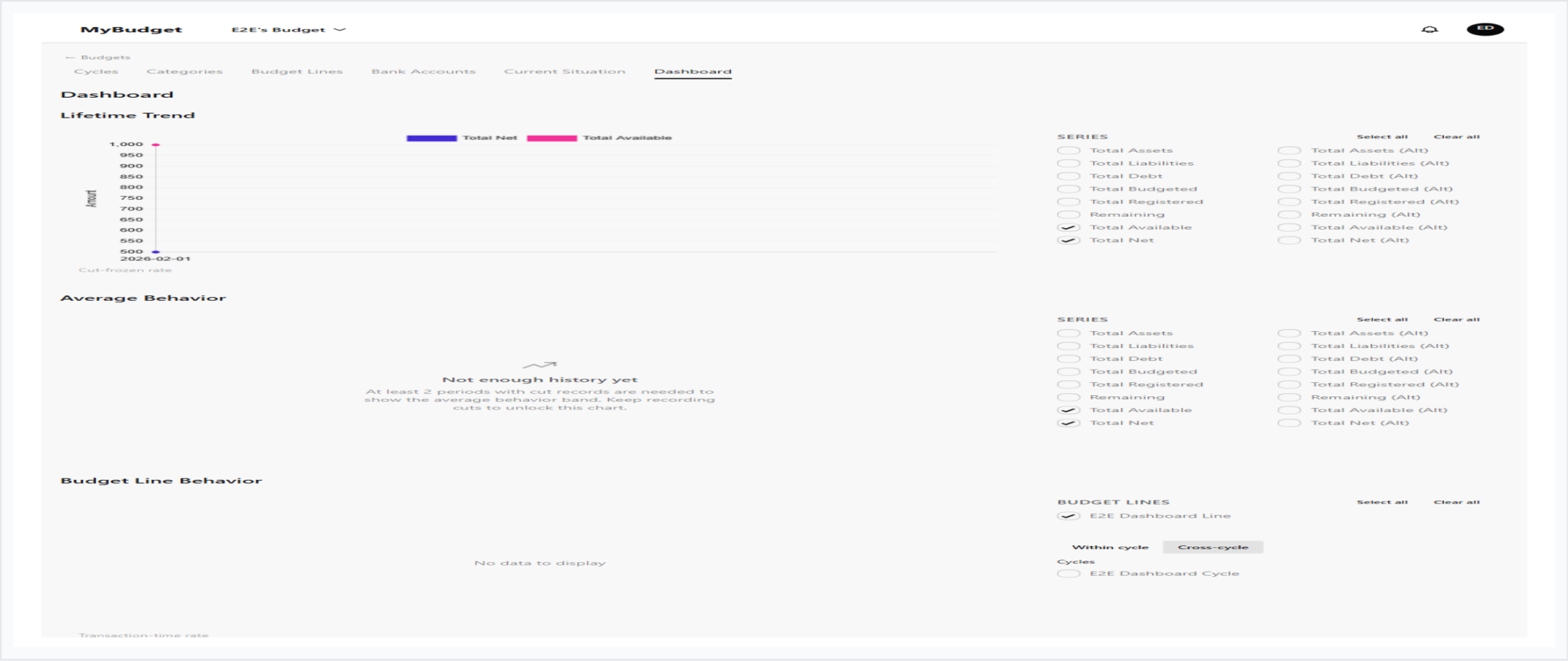


Comparación de rubros — seleccionado

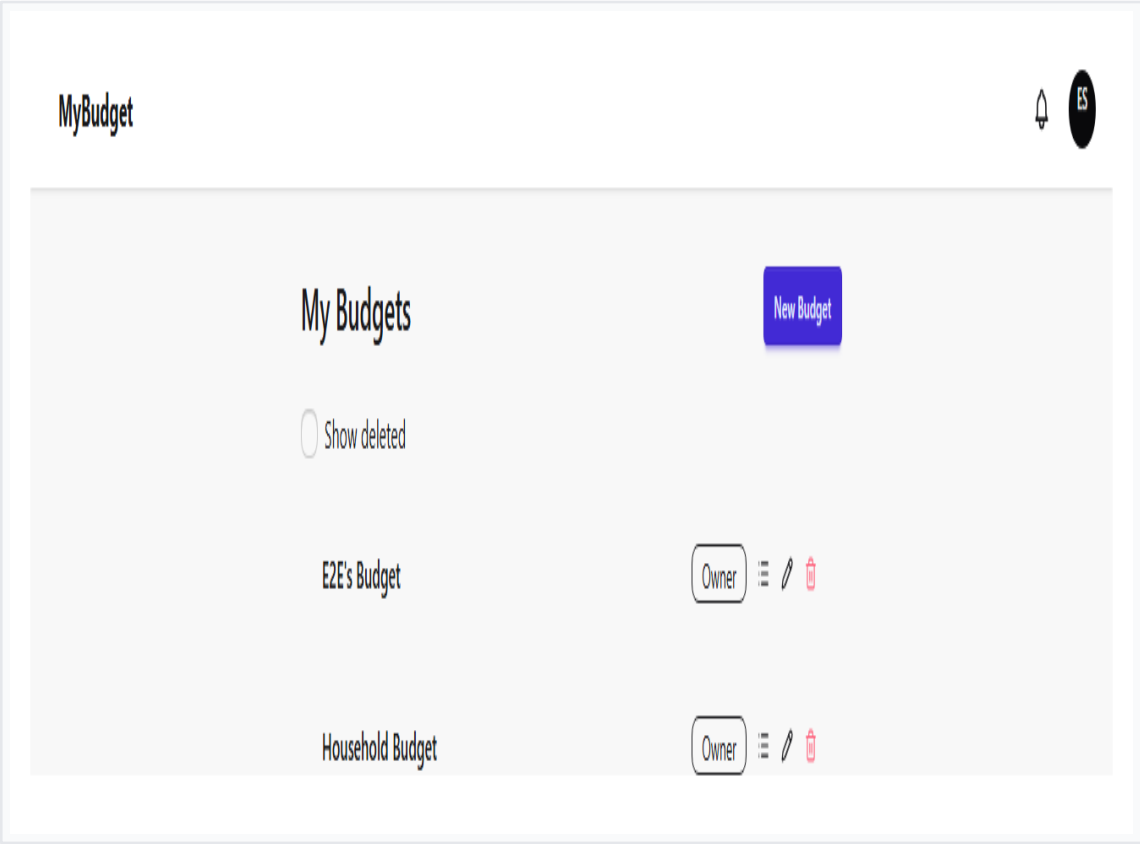


Rubro y 2 períodos seleccionados — comparación período a período dentro del ciclo.

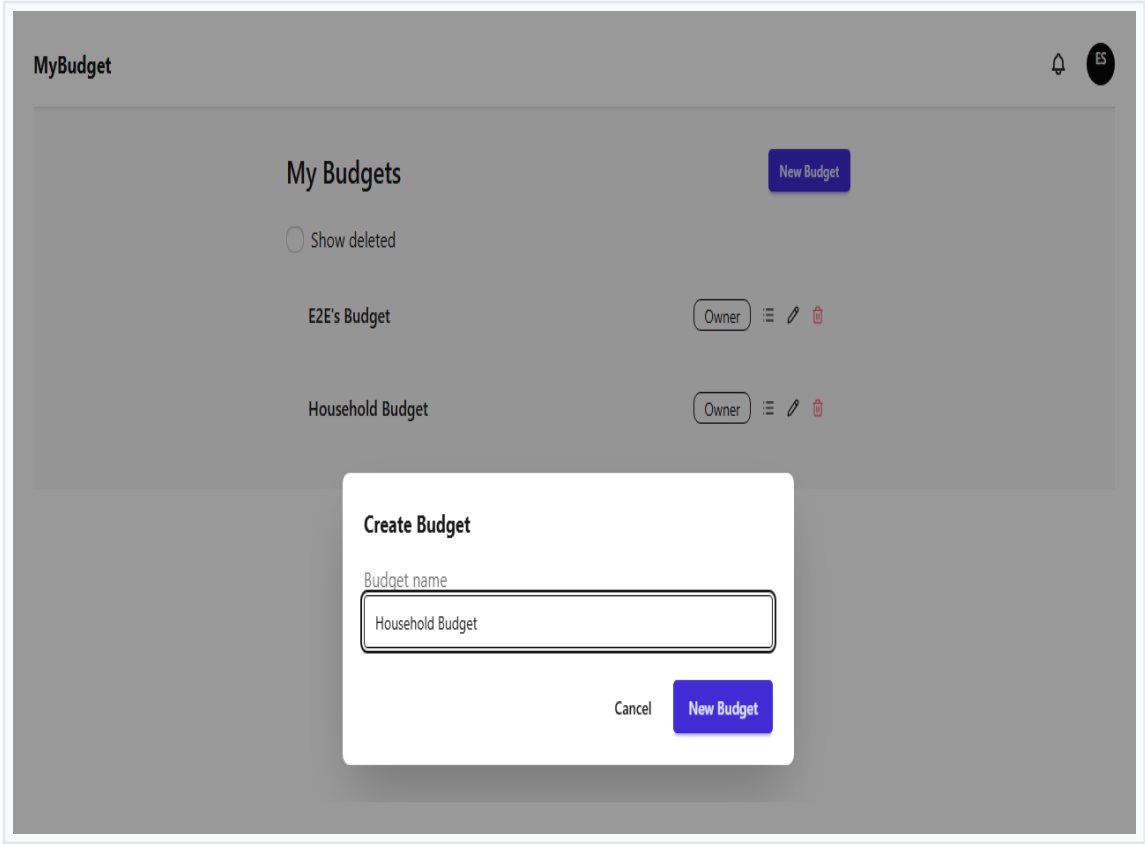
Modo entre ciclos



Presupuestos — lista y creación

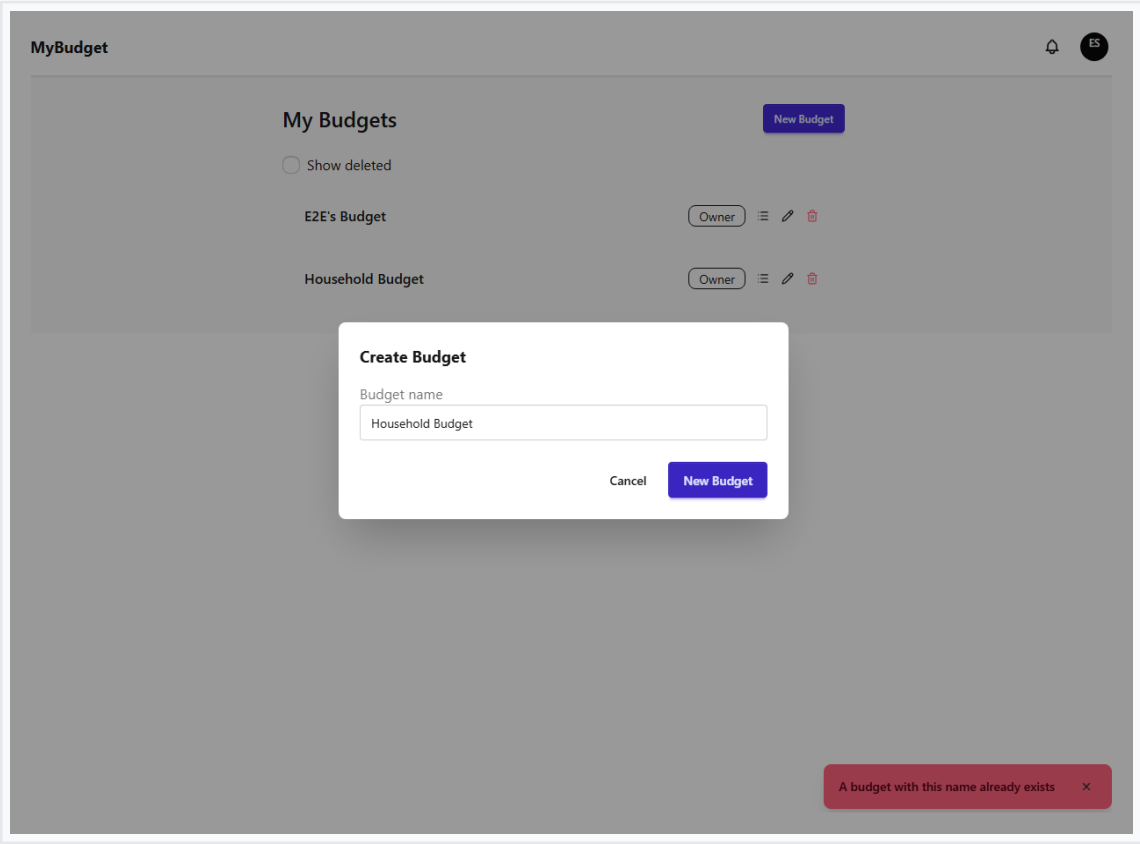


Lista de presupuestos

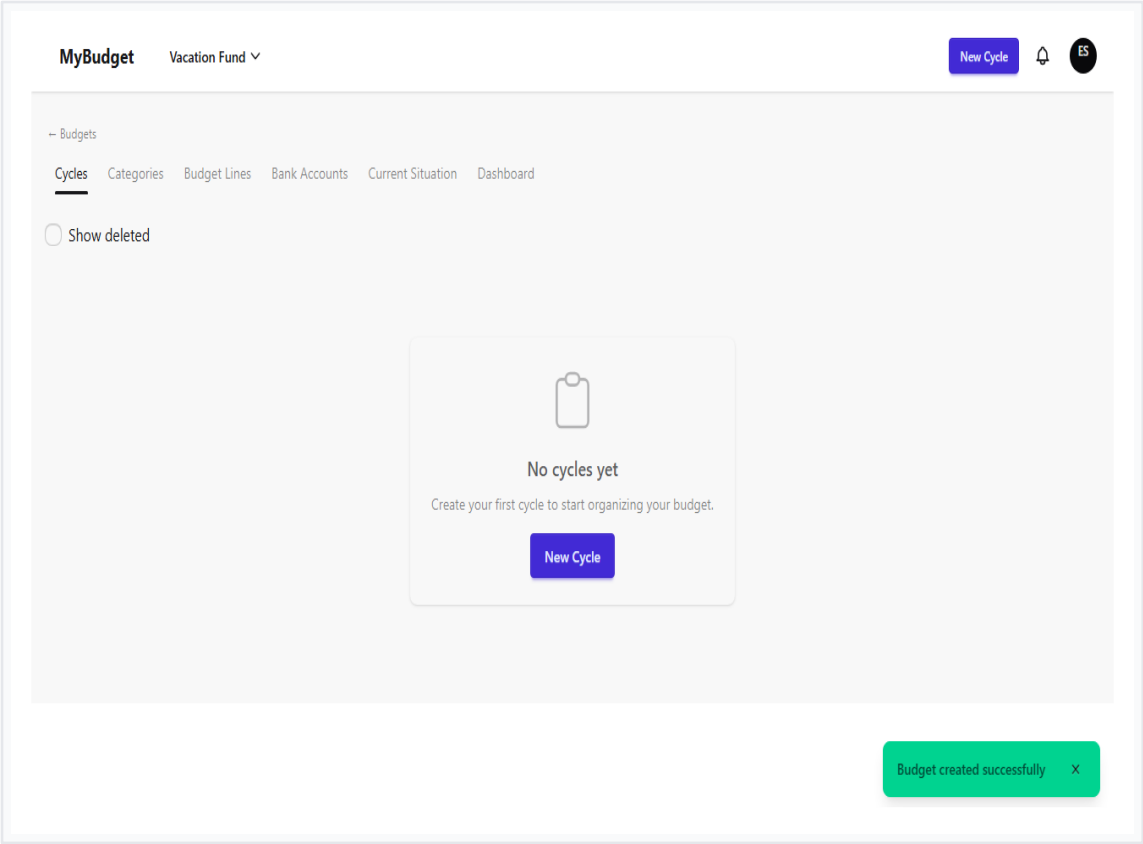


Formulario de creación

Presupuestos — validación y éxito

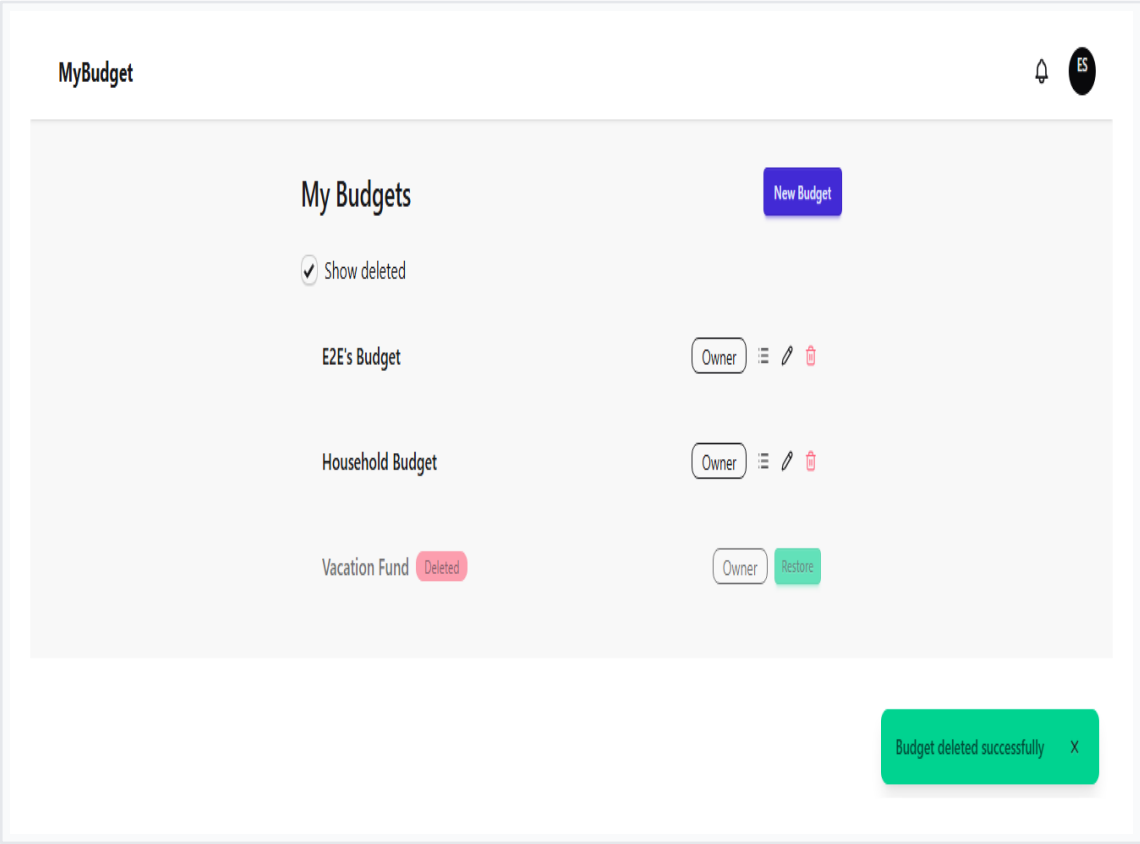


Error: nombre duplicado

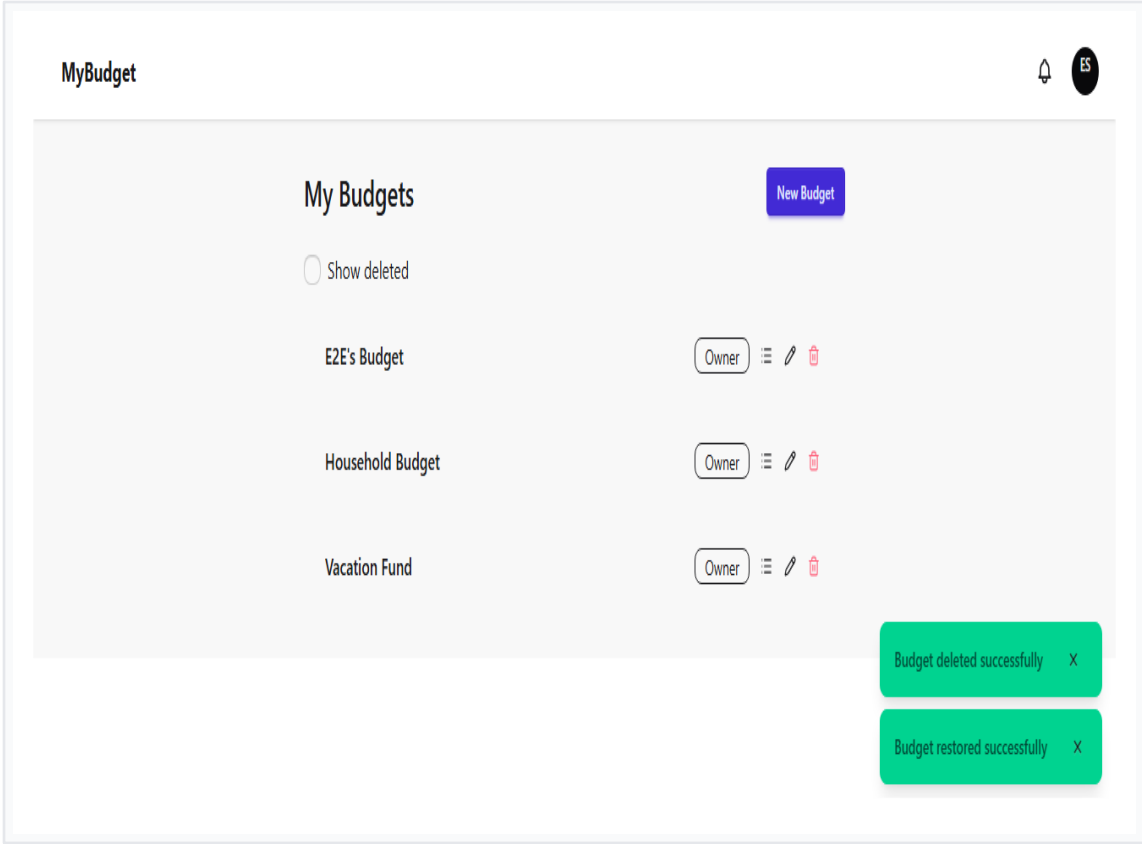


Creación exitosa

Presupuestos — eliminados y restauración

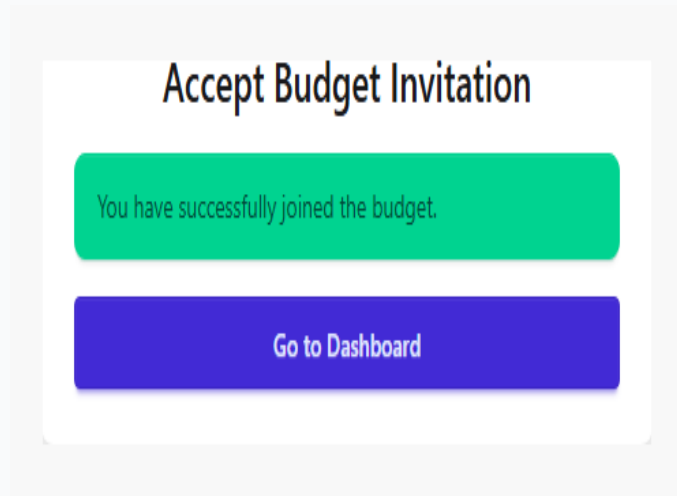


Mostrar eliminados

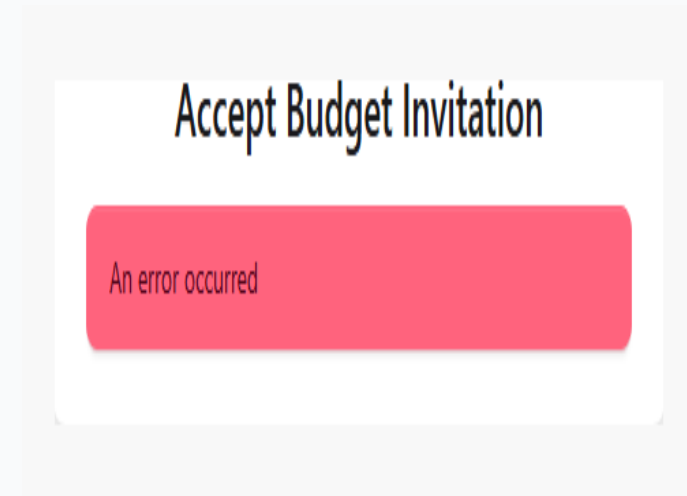


Restauración exitosa

Invitaciones — resultado



Invitación aceptada



Token inválido

Pruebas automatizadas

Capa	Cifras (ejemplo: cambio dashboard)
Backend — unitarias	488 pruebas (xUnit + NSubstitute + Shouldly)
Backend — integración	275/278 pruebas (Postgres real vía WebApplicationFactory)
Frontend — unitarias	619 pruebas (Vitest + Testing Library)
End-to-end	115 pruebas (Playwright, navegador real)

Complemento cualitativo: TDD estricto en todo el ciclo, 0 issues CRITICAL en las verificaciones finales de cada uno de los 23 cambios. Fuente: [openspec/ROADMAP.md](#).

Observabilidad y despliegue

Observabilidad

- Logs estructurados: Serilog + Seq.
- Trazas distribuidas: OpenTelemetry + Jaeger.
- Auditoría de mutaciones (AuditLog) y eventos de seguridad (SecurityAuditLog), con retención de 90 días.

Despliegue en producción

- Hetzner (VPS) + Caddy (reverse proxy / TLS automático) + Brevo (correo transaccional real).
- URL pública: mybudget-aras.duckdns.org
- Lecciones aprendidas documentadas en [docs/Deployment-LessonLearned.md](#).

Seguridad y control de acceso

- RBAC de 4 niveles: Owner, Admin, Operator, Read-only — por presupuesto.
- JWT de corta duración (15 min) + refresh token rotativo (7 días).
- Bloqueo de cuenta tras 5 intentos fallidos de login (30 min).
- Historial de las últimas 5 contraseñas — no se pueden reutilizar.

Estado y alcance futuro (MVP B)

- MVP A completo: 23 cambios shippeados y archivados.
- MVP B planeado (no iniciado): proyectos, compromisos financieros, cuotas/deudas, import/export.
- Alcance controlado deliberadamente para este TFM — MVP B queda como trabajo futuro.

Gracias

Repositorio: <https://github.com/arasman/MyBudget>

App en vivo: mybudget-aras.duckdns.org

Contacto: arasman@gmail.com